

ADAPTIVE MULTIMODAL STEGANOGRAPHY FOR SECURING MEDICAL HEALTH DATA

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled “**ADAPTIVE MULTIMODAL STEGANOGRAPHY FOR SECURING MEDICAL HEALTH DATA**” is the bonafide work of **MAHIMA S. (2022115018), KAVITHA E. (2022115022), DHANALAKSHMI G. (2022115038) and YOGANISHTA M. (2022115066)** who carried out project work under my supervision. Certified further that to the best of my knowledge and belief, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or an award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Medical data, including patient records, reports, images, and multimedia content, contains highly sensitive information that requires secure storage and transmission. While cryptographic techniques ensure confidentiality, the presence of encrypted data may still reveal sensitive information. Steganography addresses this by hiding data within multimedia carriers, but traditional single-modal approaches lack robustness and flexibility. Additionally, medical data is heterogeneous, consisting of text, images, and audio, making single-modal methods less effective. This project proposes an adaptive multimodal steganography approach to securely embed medical data across multiple carriers, improving security, imperceptibility, and robustness.

This project proposes an adaptive multimodal steganography framework to enhance the security of sensitive medical data by utilizing multiple multimedia carriers. The system supports various embedding techniques, including text-in-audio, text-in-video, image-in-image, and audio-in-image, allowing different types of medical information to be embedded into the most suitable carrier modality. By intelligently mapping each data type to an appropriate medium, the framework ensures efficient utilization of carrier capacity while preserving the visual and perceptual quality of the cover media. Unlike traditional single-modal approaches, the proposed multimodal strategy distributes secret data across multiple carriers, reducing payload concentration and minimizing the risk of detection and steganalysis attacks. At the receiver end, the system enables accurate extraction and reconstruction of the original medical data, ensuring data integrity and reliability. Experimental evaluation demonstrates that the proposed system achieves approximately 95% data recovery while maintaining high imperceptibility and robustness against common distortions, making it a secure and efficient solution for safeguarding healthcare information.

ABSTRACT (TAMIL)

இந்த திட்டம் மருத்துவ தரவுகளின் பாதுகாப்பை மேம்படுத்துவதற்காக 'அடாப்டிவ் மல்டிமோடல் ஸ்டிகனோகிராபி' அடிப்படையிலான ஒரு முன்னேற்றமான கட்டமைப்பை முன்வைக்கிறது. நோயாளர் பதிவுகள், ஆய்வக அறிக்கைகள், மருத்துவ படங்கள் மற்றும் ஒலி/வீடியோ போன்ற தகவல்கள் மிகவும் ரகசியமானவை என்பதால், அவற்றை பாதுகாப்பாக சேமித்து பரிமாறுதல் அவசியமாகிறது. பாரம்பரிய குறியாக்க முறைகள் தரவின் உள்ளடக்கத்தை பாதுகாக்க உதவினாலும், குறியாக்கப்பட்ட தரவு இருப்பது itself சந்தேகத்தை ஏற்படுத்தும். இதனை சமாளிக்க ஸ்டிகனோகிராபி முறைகள் பயன்படுத்தப்படுகின்றன, ஆனால் single-modal அணுகுமுறைகள் பல்வேறு தரவுகளை கையாளுவதில் திறனற்றவையாகவும், பாதுகாப்பு மற்றும் நெகிழ்வுத்தன்மை குறைவாகவும் உள்ளன. மருத்துவ தரவுகள் heterogeneous ஆக இருப்பதால் (உரை, படம், ஒலி), ஒவ்வொரு தரவிற்கும் வேறுபட்ட அணுகுமுறை தேவையாகிறது.

இந்த திட்டத்தில், படம், ஒலி மற்றும் வீடியோ போன்ற பல்வேறு ஊடகங்களை பயன்படுத்தி, உரை-ஒலி, உரை-வீடியோ, படம்-படம், ஒலி-படம் போன்ற embedding முறைகள் செயல்படுத்தப்படுகின்றன. தரவின் தன்மை, அளவு மற்றும் carrier திறனைப் பொருத்து சரியான ஊடகம் தானாகத் தேர்வு செய்யப்படும் adaptive முறையால், பாதுகாப்பு மட்டுமின்றி carrier பயன்பாடும் மேம்படுகிறது. மேலும், ரகசிய தரவு பல carrier-களில் பிரித்து சேமிக்கப்படுவதால், payload concentration குறைந்து, steganalysis தாக்குதல்களுக்கு எதிரான பாதுகாப்பு அதிகரிக்கிறது. பெறுபவர் பக்கம், மறைக்கப்பட்ட தகவல் துல்லியமாக மீட்டெடுக்கப்பட்டு, அதன் முழுமைத்தன்மை மற்றும் நம்பகத்தன்மை உறுதி செய்யப்படுகிறது. AES போன்ற குறியாக்க முறைகள் இணைக்கப்பட்டுள்ளதால் கூடுதல் பாதுகாப்பு கிடைக்கிறது. பரிசோதனை முடிவுகள் சுமார் 95% தர மீட்பு திறனை மற்றும் உயர்ந்த imperceptibility, robustness ஆகியவற்றை காட்டுகின்றன. எனவே, இந்த அமைப்பு மருத்துவத் துறை, கிளவுட் சேமிப்பு மற்றும் IoMT சூழல்களில் பாதுகாப்பான தர பரிமாற்றத்திற்கான ஒரு நம்பகமான தீர்வாக விளங்குகிறது.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ARSS	Analysis and Resynthesis Sound Spectrograph
CFSTC	Cost Function-driven Syndrome Trellis Code
CNN	Convolutional Neural Network
DCT	Discrete Cosine Transform
DFT	Discrete Fourier Transform
DWT	Discrete Wavelet Transform
GAN	Generative Adversarial Network
HEVC	High Efficiency Video Coding
INN	Invertible Neural Network
LSB	Least Significant Bit
MFICF	Multi-Factor-Involved Cost Function
ML	Machine Learning
MSE	Mean Square Error
PSNR	Peak Signal-to-Noise Ratio
PUPM	Prediction Unit Partition Mode
SNR	Signal-to-Noise Ratio
SSIM	Structural Similarity Index Measure

CHAPTER 1

INTRODUCTION

Medical data is one of the most sensitive and critical forms of information in modern digital systems, encompassing patient records, diagnostic reports, laboratory results, prescriptions, medical images, and other clinical data. Ensuring its confidentiality, integrity, and availability is essential, as unauthorized access can lead to privacy violations and misuse. With the increasing adoption of electronic health records and digital healthcare platforms, medical data is now widely stored, processed, and transmitted in digital form. Unlike conventional data, it is inherently heterogeneous, including structured text, high-resolution images, and sometimes audio or video, each with distinct characteristics and security requirements. This diversity makes protecting medical data particularly challenging, especially during transmission and cloud storage, highlighting the need for robust security mechanisms to preserve patient privacy, maintain trust, and comply with regulatory standards.

1.1 OVERVIEW OF STEGANOGRAPHY

Steganography is the art and science of hiding secret information within a medium such that the existence of the message is concealed from unintended observers. Unlike cryptography, which scrambles data to make it unreadable, steganography focuses on concealment, allowing secret communication to occur without attracting attention.

Common media for steganography include images, audio, video, text, and even network protocols. In digital steganography, secret data are embedded into digital files using specialized algorithms that modify the underlying medium imperceptibly, ensuring the hidden information remains undetectable under normal inspection.

Steganography finds applications in secure communication, copyright protection, digital watermarking, and data authentication. With the rise of multimedia

content and online data sharing, steganography provides a subtle yet effective means to ensure privacy in everyday digital interactions.

1.2 COVERED STEGANOGRAPHY

Cover steganography, commonly referred to as traditional steganography, is a data hiding technique in which secret information is embedded within a carrier medium such as an image, audio file, video, or text document. The primary objective of this method is to conceal the existence of the hidden data by making subtle modifications to the cover object. These modifications are designed to be imperceptible to human perception while enabling the reliable extraction of the secret information at the receiver's end.

Traditional steganographic techniques typically operate by altering specific components of the carrier, such as pixel values in images, frequency coefficients in transform-domain methods, or signal amplitudes in audio files. Although effective for covert communication, modifying the cover medium may introduce statistical irregularities that can be detected by steganalysis tools. Despite this limitation, cover steganography remains widely used due to its simplicity, flexibility, and relatively high data embedding capacity.

1.3 MULTIMODAL STEGANOGRAPHY

Multimodal steganography extends traditional techniques by integrating multiple media types or channels for data hiding. For instance, text, images, audio, and video can be combined to encode or distribute secret messages, thereby increasing the capacity, security, and robustness of the system.

In multimodal steganography, messages are mapped across multiple data modalities or databases, making unauthorized extraction significantly more difficult. This approach also enables cross-verification mechanisms, where messages can be validated across different channels to enhance reliability. Furthermore, multimodal

techniques provide scalability and flexibility, facilitating secure communication in complex and heterogeneous environments.

1.4 SECURITY ISSUES IN MEDICAL DATA

Medical data security has become a major concern due to the increasing digitization of healthcare systems and the growing exchange of sensitive information across distributed networks. Patient records, diagnostic reports, medical images, and biomedical signals are frequently transmitted through hospital information systems, cloud platforms, and telemedicine environments, making them vulnerable to interception, unauthorized access, tampering, and data leakage.

Unlike conventional digital information, medical data requires not only confidentiality but also preservation of integrity and quality. Even minor alterations in diagnostic content may affect clinical interpretation and decision making. In addition, the heterogeneous nature of medical data introduces challenges in applying a single protection mechanism to all modalities.

Traditional security mechanisms such as encryption protect confidentiality, but they do not conceal the presence of sensitive information and may still attract attention during transmission. These limitations highlight the need for additional protection mechanisms capable of securing medical data while preserving quality and reducing exposure risks.

1.5 NEED FOR ADAPTIVE MULTIMODAL STEGANOGRAPHY IN MEDICAL DATA

Conventional single-modal steganography systems embed secret information within a single carrier, often leading to payload concentration, reduced imperceptibility, and increased vulnerability to detection. Although multimodal steganography improves security by distributing data across multiple carriers, many existing approaches rely on fixed embedding assignments and lack intelligent adaptation.

Medical data consists of heterogeneous modalities such as text, diagnostic images, and biomedical audio, each having different embedding requirements and cover suitability. These characteristics necessitate a framework capable of adaptively selecting suitable carriers according to data type, payload size, and embedding constraints.

Adaptive multimodal steganography addresses this requirement through decision-driven carrier selection and distributed embedding. By intelligently mapping secret data to appropriate carriers, the approach improves embedding efficiency, reduces detectability, enhances robustness, and supports reliable recovery. These advantages justify the need for adaptive multimodal steganography in securing medical data.

1.6 PROBLEM STATEMENT

Medical data, including patient records, diagnostic reports, medical images, and associated multimedia content, contains highly sensitive information that must be securely protected during storage and transmission. Although cryptographic techniques ensure data confidentiality, the visible presence of encrypted files may still reveal the existence of sensitive information, potentially attracting unwanted attention. Steganography offers an additional layer of protection by concealing secret data within multimedia carriers. However, conventional steganographic approaches primarily rely on single-modal embedding, where all sensitive data is hidden within a single carrier medium.

Single-modal steganography introduces several limitations, including increased payload density, reduced imperceptibility, and vulnerability to detection or carrier distortion. Furthermore, medical data is inherently heterogeneous, comprising text, images, and audio, each requiring different embedding capacities and security characteristics. Concentrating diverse data types within a single medium reduces flexibility and creates a single point of compromise. These challenges highlight the need for a multimodal steganography framework capable of distributing medical data across multiple complementary carriers to enhance robustness, security, and resistance to steganalysis.

1.7 RESEARCH CONTRIBUTIONS

The major contributions of this work are summarized as follows:

- Development of an adaptive multimodal steganography framework for heterogeneous medical data.
- Dynamic carrier selection based on payload type, embedding suitability, and carrier capacity.
- Integration of multiple embedding mechanisms including text-in-video, text-in-audio, image-in-image, and audio-in-image steganography.
- A distributed embedding strategy to reduce payload concentration and improve resistance to steganalysis.
- Reliable extraction and reconstruction framework for secure recovery of concealed medical information.

1.8 ORGANIZATION OF THE THESIS

The thesis is organized into six chapters as follows:

- **Chapter 1: Introduction** — Provides an overview of steganography, multimodal data hiding, medical data security issues, and the motivation for adaptive multimodal steganography.
- **Chapter 2: Literature Survey** — Reviews existing steganography methods, their drawbacks, and recent improvements.
- **Chapter 3: System Architecture** — Provides a detailed description of the proposed architecture and its working modules.

- **Chapter 4: System Implementation** — Describes the implementation of adaptive carrier selection, multimodal embedding modules, extraction mechanisms, and communication workflow.
- **Chapter 5: Results and Analysis** — Presents the experimental results, performance evaluation.
- **Chapter 6: Conclusions and Future Work** — Summarizes the work done, key findings, and possible future research directions.

CHAPTER 2

LITERATURE SURVEY

2.1 CONCEPT OF STEGANOGRAPHY

Steganography is a fundamental technique in information security that focuses on concealing the existence of secret communication within digital media. It has evolved from basic embedding methods to advanced intelligent systems to improve security, capacity, and robustness.

2.1.1 Fundamentals and Applications of Steganography

Steganography is a technique used to hide secret information within digital media such that the existence of the message remains concealed, as described by **Subramanian et al.** [1]. Traditional cover-based methods embed secret bits by modifying carrier media such as images, audio, video, or text. Common techniques include spatial domain approaches like Least Significant Bit and transform domain methods such as Discrete Cosine Transform and Discrete Wavelet Transform. These approaches aim to preserve perceptual quality while hiding confidential data. However, even slight modifications may introduce statistical artifacts that can be detected. This limitation has motivated the development of more secure and less detectable embedding strategies.

Steganography is defined as a method of concealing secret communication by embedding data within a carrier medium, as explained by **Ali et al.** [2]. The technique supports various media types including images, audio, video, and text based on application requirements. Its primary objective is to hide the existence of the message while maintaining the original quality of the carrier. It is considered a major branch of information hiding and plays an important role in secure communication systems. Methods can be categorized based on carrier type, embedding domain, and

security level. Applications include secure communication, data protection, and digital watermarking.

2.1.2 Statistical Steganalysis Using RS Analysis

A statistical steganalysis method based on RS analysis was proposed by **Fridrich et al.** [3]. The method analyzes pixel groups and classifies them into regular and singular sets to detect embedding artifacts. It demonstrates that even minimal LSB modifications can introduce detectable statistical anomalies. The approach is highly effective against simple spatial domain techniques. It laid the foundation for many modern steganalysis methods. The study highlights the vulnerability of traditional embedding techniques.

2.2 TRADITIONAL EMBEDDING METHODS

Traditional embedding methods form the foundation of steganography by modifying carrier media to hide secret data. These techniques are broadly classified into spatial-domain and transform-domain approaches based on how data is embedded.

2.2.1 Spatial Domain Techniques

A widely used spatial domain embedding technique was analyzed by **Cheddad et al.** [4]. The method performs data hiding by modifying the least significant bits of pixel values. It provides high embedding capacity with minimal perceptual distortion. The technique is simple and computationally efficient to implement. However, it is highly vulnerable to statistical detection and common attacks. The approach lacks robustness under compression and noise, making it suitable mainly for low-security applications.

2.2.2 Transform Domain Techniques

A transform domain technique using Discrete Cosine Transform was explored by **Cox et al.** [5]. The system embeds data into frequency coefficients rather than directly modifying pixel values. This improves robustness against compression and filtering operations. It also provides better imperceptibility compared to spatial domain techniques. However, computational complexity is higher and embedding capacity is limited. The study shows that transform domain approaches offer improved security compared to basic methods.

The evolution of image steganography from traditional methods to modern deep learning approaches was discussed by **Huang et al.** [6]. Classical techniques include spatial-domain methods such as Least Significant Bit substitution and pixel value differencing. Transform-domain approaches such as Discrete Cosine Transform, Discrete Wavelet Transform, and Discrete Fourier Transform are also widely used. These methods aim to achieve high imperceptibility, embedding capacity, and robustness while preserving image quality. Recent advancements have introduced deep learning techniques based on convolutional neural networks and generative adversarial networks. However, challenges remain in balancing capacity, robustness, and imperceptibility under various attacks such as compression, noise, and image processing operations.

2.3 IMAGE STEGANOGRAPHY

Image steganography focuses on hiding secret information within digital images while preserving visual quality and minimizing detectability. Recent advancements have significantly improved embedding capacity and robustness through deep learning-based approaches.

2.3.1 LiDiNet Based Image Steganography

A lightweight deep learning-based steganography system called LiDiNet

was proposed by **Li et al.** [7]. The method hides a full-size secret image within a cover image while minimizing contour shadows and color distortion. It utilizes invertible neural networks with affine coupling layers to enable efficient embedding and extraction. Attention modules enhance feature fusion and improve reconstruction quality. The architecture ensures stable training with high embedding capacity and robustness against steganalysis. Experimental results demonstrate superior visual quality and reconstruction accuracy compared to existing methods.

2.3.2 Encoder–Decoder Based Image Steganography

A deep learning-based image steganography system using an encoder–decoder architecture was proposed by **Zhang et al.** [8]. The encoder embeds the secret image into the cover image while the decoder reconstructs it at the receiver end. This end-to-end framework improves embedding efficiency and reduces visible distortion. The model maintains high visual quality of the stego image. It enhances robustness against operations such as compression, cropping, and noise addition. Experimental results show improved embedding capacity and better recovery performance compared to traditional techniques.

2.3.3 End-to-End Neural Image Steganography

A deep learning-based steganography system was proposed by **Baluja** [9]. The approach uses neural networks to hide one image within another. It consists of preparation, hiding, and reveal networks for end-to-end learning. The method achieves high embedding capacity while preserving visual quality. However, the training process is computationally intensive. The study demonstrates the effectiveness of deep learning for secure image data hiding.

2.4 VIDEO STEGANOGRAPHY

Video steganography focuses on hiding secret information within video streams while maintaining temporal consistency and visual quality. Modern approaches

primarily exploit compression structures, motion information, and deep generative models to improve embedding efficiency and robustness.

2.4.1 Compression Domain Based Video Steganography

An adaptive video steganography system based on Prediction Unit Partition Mode was proposed by **Yu et al.** [10]. The method targets High Efficiency Video Coding streams to balance embedding capacity, efficiency, and security. It addresses the limitations of fixed partition modes such as $2N \times 2N$ that introduce predictable patterns. A Multi-Factor-Involved Cost Function evaluates embedding decisions based on distortion, bitrate variation, and statistical detectability. A 7-ary notational system improves embedding efficiency across different partition modes. Experimental results show improved efficiency, reduced bitrate increase, and enhanced resistance to steganalysis.

A multi-domain video steganography system based on HEVC coding structure was proposed by **Li et al.** [11]. The method utilizes Transform Unit partitioning and Intra Prediction Modes for adaptive data embedding. It selects embedding regions based on coding characteristics to minimize distortion. A distortion optimization framework balances embedding capacity and visual quality. The system achieves higher capacity compared to single-domain approaches. Experimental results demonstrate improved efficiency and imperceptibility in HEVC-based video steganography.

2.4.2 Motion Vector Based Video Steganography

A video steganography system based on motion vectors was proposed by **Cao et al.** [12]. The method embeds secret data into motion vectors of compressed video streams. It preserves visual quality while maintaining compatibility with compression standards. The approach demonstrates robustness against compression and common video processing operations. It supports real-time applications due to efficient

embedding. However, embedding capacity remains moderate compared to advanced techniques.

2.4.3 Generative Video Steganography Using RoGVS

A robust generative video steganography framework called RoGVS was proposed by **Mao et al.** [13]. The method embeds secret information within semantic features instead of modifying pixel values directly. It improves robustness against distortions such as compression, resizing, and format conversion. An end-to-end generative network is used with an adaptive message embedding module. The system maintains high visual quality while increasing embedding capacity. Experimental results confirm improved robustness, imperceptibility, and reconstruction performance.

2.5 AUDIO STEGANOGRAPHY

Audio steganography focuses on hiding secret information within audio signals while preserving perceptual quality and minimizing audible distortion. Modern approaches enhance robustness by leveraging spectrogram representations and deep learning-based encoder–decoder architectures.

2.5.1 Audio In Image Steganography Using ARSS

A novel audio-in-image steganography system was proposed by **Krishnan et al.** [14]. The method uses the Analysis and Resynthesis Sound Spectrogram to convert audio signals into spectrogram representations. These spectrograms are embedded into high-resolution images using a multi-channel non-lossy embedding approach. The technique requires minimal preprocessing and preserves the quality of both the cover image and hidden audio data. It ensures efficient data storage while maintaining high imperceptibility and security. Experimental results demonstrate improved robustness and reduced computational complexity compared to conventional techniques.

2.5.2 Encoder–Decoder Based Audio Steganography

A high-fidelity audio steganography system based on an encoder–decoder architecture was proposed by **Zhang et al.** [15]. The method separates audio signals into content and style components using an encoder network. Secret information is embedded into the content vector while preserving perceptual features such as timbre and tone. The decoder reconstructs the stego audio by combining the modified content vector with the original style features. This approach avoids traditional time-domain and frequency-domain embedding techniques. Experimental results demonstrate enhanced embedding capacity, robustness, and resistance to steganalysis.

2.5.3 LSB Based Audio Steganography

Least Significant Bit-based data hiding techniques are widely used in audio steganography due to their simplicity and high embedding capacity. These approaches are inspired by spatial domain embedding concepts widely studied in image steganography, including foundational works by **Wu and Tsai** [16]. The technique embeds secret information into the least significant bits of audio samples with minimal perceptual distortion. It preserves audio quality effectively under ideal conditions. However, the method is highly sensitive to noise and compression, which can easily destroy hidden data. Therefore, LSB-based approaches are generally suitable only for low-security applications.

2.6 ATTACK RELATED WORK

Steganalysis attacks focus on detecting or removing hidden information embedded within digital media such as images and videos. These attacks are broadly categorized into active removal attacks and detection-based statistical or learning-based methods.

2.6.1 Erase And Repair Based Steganalysis Attack

A box-free removal attack for deep steganography was proposed by **Liu et al.** [17]. The method actively removes hidden images without requiring prior knowledge of the embedding scheme. It introduces an erase-and-repair framework that eliminates secret content while preserving visual quality. The approach targets deep learning-based steganography by identifying embedded information within container images. It effectively suppresses the recovery of hidden images across multiple deep hiding models. Experimental results demonstrate strong performance in disrupting hidden data recovery across various datasets.

2.6.2 Deep Learning Based Steganalysis

A machine learning-based detection system was proposed by **Fridrich and Kodovsky** [18]. The method utilizes rich models for extracting high-dimensional features from images. It significantly improves classification accuracy for detecting steganographic content. The system is capable of identifying complex embedding techniques. However, feature extraction and computation are resource-intensive. The study represents a major advancement in feature-based steganalysis.

A convolutional neural network-based detection system was proposed by **Ye et al.** [19]. The model automatically learns discriminative features for steganalysis. It achieves higher detection accuracy compared to traditional machine learning approaches. Manual feature engineering is eliminated through end-to-end learning. The system effectively handles complex and adaptive embedding techniques. However, large datasets and extensive training are required for optimal performance.

An efficient deep learning-based steganalysis system was proposed by **Yedroudj et al.** [20]. The model introduces a specialized convolutional neural network architecture for spatial image steganalysis. It incorporates preprocessing filters and optimized convolution layers to enhance feature extraction. The system achieves improved detection accuracy compared to traditional approaches. It effectively

captures subtle statistical differences between cover and stego images. However, careful parameter tuning and training are required for optimal performance.

2.7 ADAPTIVE STEGANOGRAPHY

Adaptive steganography focuses on dynamically selecting embedding locations based on image content or learned policies to improve security and imperceptibility. These methods reduce detectability by avoiding uniform embedding patterns and adapting to structural characteristics of the cover media.

2.7.1 Adaptive Steganography Using CNN

An adaptive embedding system was proposed by **Tang et al.** [21]. The method utilizes convolutional neural networks to dynamically select suitable embedding regions within images. It reduces statistical artifacts that can be detected by steganalysis techniques. The system improves imperceptibility by embedding data into complex texture regions. It adapts to image content to enhance overall security and robustness. Experimental results demonstrate improved resistance against modern detection methods.

2.7.2 Reinforcement Learning Based Steganography

An intelligent adaptive steganographic system was proposed by **Yang et al.** [22]. The approach integrates reinforcement learning with generative adversarial networks for optimized embedding. It learns optimal embedding strategies by interacting with the environment and minimizing detection risk. The system reduces detectability by dynamically selecting secure embedding locations. Robustness against steganalysis attacks is significantly improved. However, the method requires high computational resources and large training datasets.

2.8 SUMMARY OF EXISTING WORK

- Existing steganography techniques demonstrate strong capability in securely hiding information across multiple media formats such as images, audio, and video.
- Traditional methods such as LSB, DCT, and DWT provide high embedding capacity but are often vulnerable to statistical and steganalysis attacks.
- Deep learning-based approaches, including CNN and GAN models, significantly improve imperceptibility and resistance to advanced detection techniques.
- Many systems maintain robustness under common transformations such as compression, resizing, noise addition, and filtering operations.
- Video and audio steganography techniques enhance robustness by embedding data in motion vectors, frequency components, and semantic features.
- Adaptive and intelligent embedding methods, including reinforcement learning-based approaches, reduce detectability by dynamically selecting embedding regions.
- Some advanced methods support reversible data hiding, which is especially useful in sensitive domains such as medical applications.
- Existing approaches achieve higher embedding capacity while maintaining acceptable visual or perceptual quality of the carrier media.
- Steganography has proven to be effective in applications such as secure communication, digital watermarking, and data protection.
- However, challenges still remain in achieving an optimal balance between embedding capacity, robustness, and imperceptibility.

2.9 OBJECTIVE OF THE PROPOSED SYSTEM

- To design a multimodal steganography framework for securing sensitive medical data.
- To implement heterogeneous embedding techniques including text in audio, text in video, image in image, and audio in image.
- To reduce detectability by avoiding high payload concentration in any single carrier medium.
- To enhance robustness against steganalysis by leveraging multiple carrier modalities.
- To ensure accurate extraction and reconstruction of hidden medical data at the receiver end.
- To maintain high imperceptibility while preserving the quality of carrier media.
- To evaluate system performance using metrics such as PSNR, SSIM, and robustness against attacks.

CHAPTER 3

SYSTEM ARCHITECTURE

The proposed Adaptive Multimodal Communication System (AMCS) is designed to protect sensitive medical data by embedding it within multimedia carriers using multiple steganographic techniques. Medical data often includes multiple types of information such as textual patient records, diagnostic images, and audio reports. Instead of embedding all information into a single carrier, the proposed system distributes the hidden data across multiple media formats such as images, audio, and video.

Figure 3.1 illustrates the overall architecture of the Adaptive Multimodal Communication System. The architecture consists of four major stages: secret data preprocessing, adaptive decision making, multimodal embedding, and extraction and reconstruction at the receiver side.

The system supports multiple steganography techniques including text-in-video, image-in-image, audio-in-image, and high-fidelity audio steganography. By using multiple carrier media, the system improves security, reduces payload concentration in a single carrier, and increases resistance to steganalysis.

3.1 SECRET DATA INPUT AND PREPROCESSING

The process begins with the input of sensitive medical information that must be securely transmitted. The input may consist of textual patient records, diagnostic images, or audio recordings depending on the application.

Before embedding, the input data undergoes preprocessing to convert it into a suitable representation for the steganographic algorithms. Textual data is converted into ASCII values and binary sequences. Image data undergoes resizing, normalization,

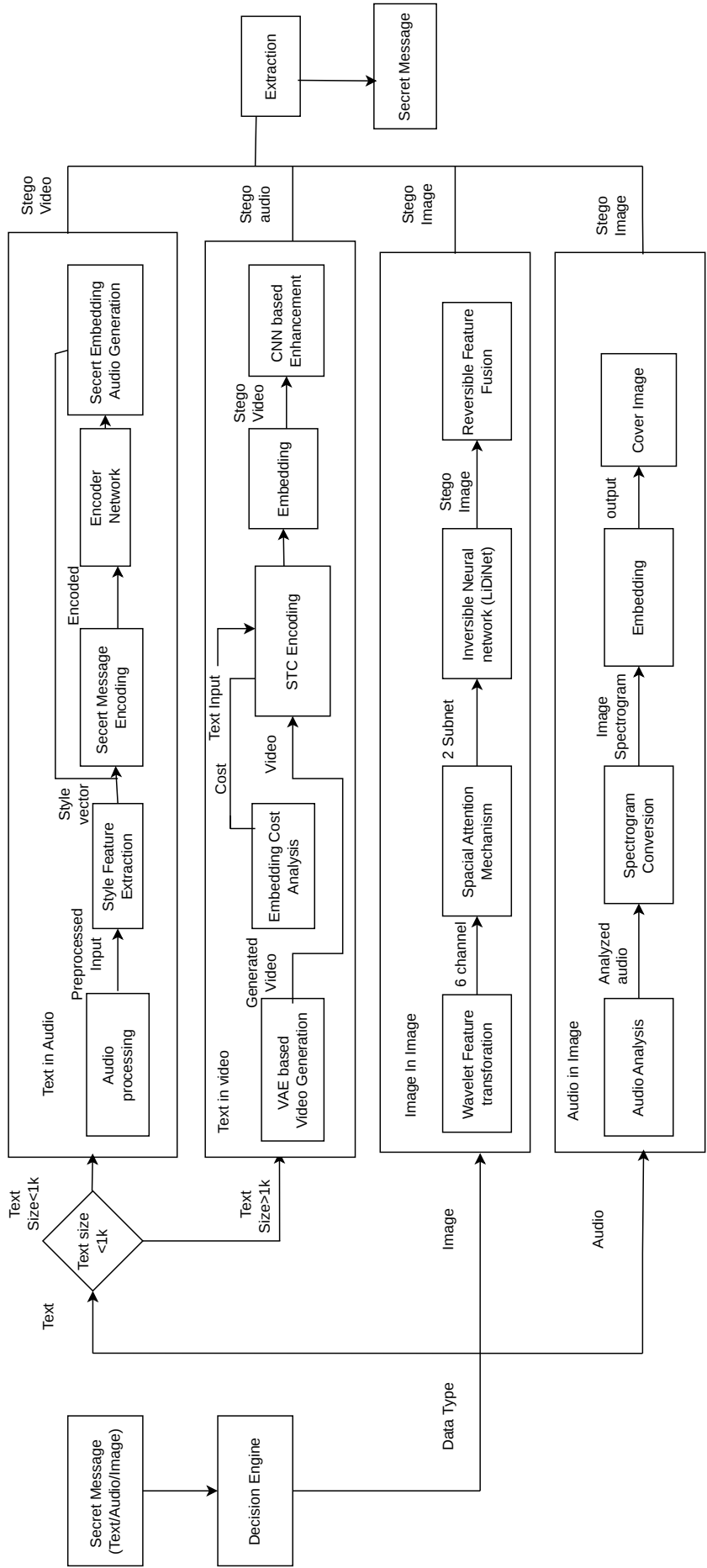


Figure 3.1: Adaptive Multimodal Communication System Architecture

and feature extraction, while audio signals are normalized and transformed into spectral representations when required.

As shown in Figure 3.1, the processed secret data is forwarded to the decision engine for selecting an appropriate embedding method.

3.2 DECISION ENGINE

The decision engine acts as the central controller of the system. Its primary function is to determine which steganographic module should be used to embed the secret data. The decision is based on input data type, available carrier media, and embedding capacity.

Based on these conditions, the decision engine selects one of the following modules:

- Text-in-Video Steganography Module
- Image-in-Image Steganography Module
- Audio-in-Image Steganography Module
- High-Fidelity Audio Steganography Module

The adaptive carrier selection performed by the decision engine in Figure 3.1 determines the embedding pipeline used by the framework.

3.3 CARRIER MEDIA SELECTION

The proposed framework employs multiple cover carriers for secret data embedding. A carrier refers to the multimedia object used to conceal hidden information while preserving perceptual quality.

The carriers used in the proposed system include:

- Audio carrier for embedding textual data
- Video carrier for large textual payloads
- Image carrier for hiding diagnostic images
- Image carrier for embedding biomedical audio represented as spectrograms

Carrier selection is adaptively determined according to payload type, cover capacity, and embedding suitability.

Table 3.1 presents the mapping between secret data modalities, carrier media, and embedding techniques used in the proposed framework.

Table 3.1: Secret Data to Carrier Mapping in AMCS

Secret Data Type	Carrier Medium	Embedding Technique
Text (Short Payload)	Audio	Text-in-Audio Steganography
Text (Large Payload)	Video	Text-in-Video Steganography
Diagnostic Image	Image	Image-in-Image Steganography
Biomedical Audio	Image	Audio-in-Image Steganography

3.4 AUDIO-IN-IMAGE STEGANOGRAPHY

The Audio-in-Image module embeds audio signals within digital images. The audio signal is first converted into a spectrogram representation using Short Time Fourier Transform (STFT).

Spectrogram values are normalized and mapped to pixel intensities within the cover image. Embedding is performed using Least Significant Bit modification while preserving the visual quality of the image.

The resulting output is a stego image containing the hidden audio signal.

3.5 IMAGE-IN-IMAGE STEGANOGRAPHY

The Image-in-Image module hides a secret image inside a cover image using an invertible neural network called LiDiNet.

Both images are transformed into feature representations using wavelet decomposition. These features are processed using spatial attention mechanisms and invertible convolution layers to preserve important spatial structures while maintaining reversibility.

Coupling blocks merge secret and cover image features. Because the operations are invertible, the hidden image can be reconstructed during extraction. The output is a stego image visually identical to the cover image while containing embedded diagnostic information.

3.6 TEXT-IN-VIDEO STEGANOGRAPHY

The Text-in-Video module hides textual medical records inside video frames. This process begins by converting input text into binary form.

A Variational Autoencoder (VAE) generates video frames with natural motion patterns. Suitable embedding regions are identified using a Modified Feature Importance Cost Function (MFICF), which determines pixels that can be modified with minimal perceptual distortion.

To reduce embedding distortion, the binary message is encoded using Syndrome Trellis Codes (STC), minimizing the number of pixel modifications. The encoded bits are then embedded using Least Significant Bit (LSB) modification, producing the final stego video.

3.7 TEXT-IN-AUDIO STEGANOGRAPHY

The text-in-Audio Steganography module embeds secret messages into audio signals while preserving audio quality. This method separates audio into content and style representations.

Secret data are inserted into the content representation while preserving style features such as pitch and tone. The modified content is then combined with original style information to reconstruct the final audio signal.

This approach maintains natural sound quality while carrying hidden information.

3.8 STEGO MEDIA TRANSMISSION

After embedding, the system generates stego media containing hidden data. Depending on the embedding method, the output may be a stego image, stego audio signal, or stego video.

As illustrated in Figure 3.1, the generated stego media is transmitted through the communication channel while appearing visually or acoustically unchanged.

3.9 EXTRACTION AND RECONSTRUCTION

At the receiver side, the extraction module retrieves the concealed information from the received stego media using the inverse process corresponding to the embedding technique selected by the decision engine. As illustrated in Figure 3.1, extraction and reconstruction represent the final stage of the Adaptive Multimodal Communication System, where hidden data are recovered and converted back into their original form.

For Audio-in-Image steganography, the embedded spectrogram information

is extracted from the stego image through inverse Least Significant Bit recovery. The recovered spectrogram is then transformed back into an audio signal using inverse Short Time Fourier Transform (ISTFT), reconstructing the original biomedical audio.

For Image-in-Image steganography, inverse transformations of the LiDiNet model are applied to separate the hidden image from the cover image. Through reversible coupling operations and inverse feature reconstruction, both the concealed diagnostic image and the original cover image are recovered.

For Text-in-Video steganography, embedded message bits are extracted from modified video frames and decoded using Syndrome Trellis Code decoding. The recovered binary sequence is converted back into textual information to reconstruct the original medical record.

For Text-in-Audio steganography, hidden data are extracted from the content representation of the stego audio signal while preserving the style components. The extracted binary information is decoded to recover the original secret message.

After extraction, the recovered outputs are reconstructed into their original modalities, including text, image, or audio, producing the final recovered medical data. This reconstruction process ensures reliable retrieval of hidden information while maintaining consistency with the adaptive multimodal embedding framework.

3.10 SUMMARY

The proposed Adaptive Multimodal Communication System integrates multiple steganographic techniques within a unified framework to securely transmit medical data. By distributing hidden information across multiple carrier media including image, audio, and video, the system improves embedding capacity, reduces detection risk, and ensures reliable recovery of concealed data at the receiver side.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 AUDIO-IN-IMAGE STEGANOGRAPHY SYSTEM

The proposed Audio-in-Image Steganography system consists of several processing stages that enable secure embedding and retrieval of audio data within a digital image. The system converts the input audio signal into a spectrogram representation and embeds it into a cover image using a Least Significant Bit (LSB) based steganography technique. The overall pipeline is illustrated in Figure 4.1.

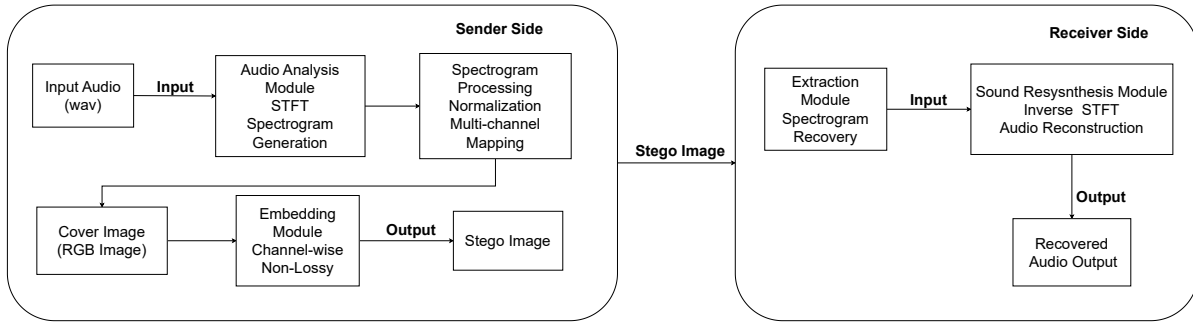


Figure 4.1: Audio-in-Image Steganography System.

4.1.1 Audio Hiding Module

The Audio Hiding Module securely embeds audio information into a digital cover image. The input audio signal is first converted into a spectrogram that captures the time–frequency characteristics of the audio. An appropriate cover image is then selected, and the spectrogram data are embedded into its pixel values using LSB substitution. Because only the least significant bit of each channel value is altered, the visual quality of the stego image remains almost unchanged.

Audio Input and Preprocessing

The system loads the input audio file and normalises the waveform to ensure compatibility with subsequent signal processing operations.

Spectrogram Generation

The preprocessed audio is transformed into a time-frequency representation using the Short-Time Fourier Transform (STFT). The resulting magnitude spectrogram is normalised and converted into an 8-bit grayscale format suitable for pixel-level embedding.

Cover Image Selection

The cover image is loaded and analysed to confirm that its resolution provides sufficient pixel capacity to store the spectrogram without noticeable distortion.

Audio Embedding

The spectrogram data are hidden in the RGB pixel values of the cover image via LSB steganography. Only the least significant bit of each channel value is altered, so the visual appearance of the image is almost unchanged.

Stego Image Generation

The modified image is saved as the stego image, which appears visually identical to the original cover image while secretly containing the embedded audio

information.

4.1.2 Audio Reconstruction Module

Audio Extraction

The hidden spectrogram data are retrieved from the stego image by reading the least significant bits of the pixel values and reversing the embedding procedure. The complete extraction procedure is formalised in Algorithm 4.1.

Algorithm 4.1 Audio Extraction from Stego Image

```

1: Input: Stego image  $S$ , stored phase matrix  $\Phi$ 
2: Output: Recovered audio signal  $A_r$ 
3: procedure AUDIOEXTRACT( $S, \Phi$ )
4:    $Spec_r \leftarrow \text{EmptyMatrix}(); \quad idx \leftarrow 0$ 
5:   for each embedding pixel location  $(r, c)$  in  $S$  do
6:      $b_R \leftarrow S[r, c, R] \text{ AND } 1$ 
7:      $b_G \leftarrow S[r, c, G] \text{ AND } 1$ 
8:      $b_B \leftarrow S[r, c, B] \text{ AND } 1$ 
9:      $Spec_r[idx] \leftarrow \text{ReconstructValue}(b_R, b_G, b_B)$ 
10:     $idx \leftarrow idx + 1$ 
11:   end for
12:    $Spec_r \leftarrow \text{Denormalise}(Spec_r)$ 
13:    $\hat{S} \leftarrow Spec_r \cdot e^{j\Phi}$   $\triangleright$  combine magnitude with stored phase
14:    $A_r \leftarrow \text{ISTFT}(\hat{S})$ 
15:   return  $A_r$ 
16: end procedure

```

Audio Reconstruction

Once the spectrogram is extracted, the Inverse Short-Time Fourier Transform (ISTFT) is applied together with the stored phase information to reconstruct the time-domain audio waveform, yielding the recovered signal A_r .

4.2 IMAGE-IN-IMAGE STEGANOGRAPHY

The LiDiNet module implements image-in-image steganography using a lightweight deep invertible neural network. Unlike conventional techniques that directly modify pixel values, LiDiNet performs feature-level transformations through invertible network blocks, enabling both embedding and extraction via forward and inverse operations. The overall architecture is illustrated in Figure 4.2.

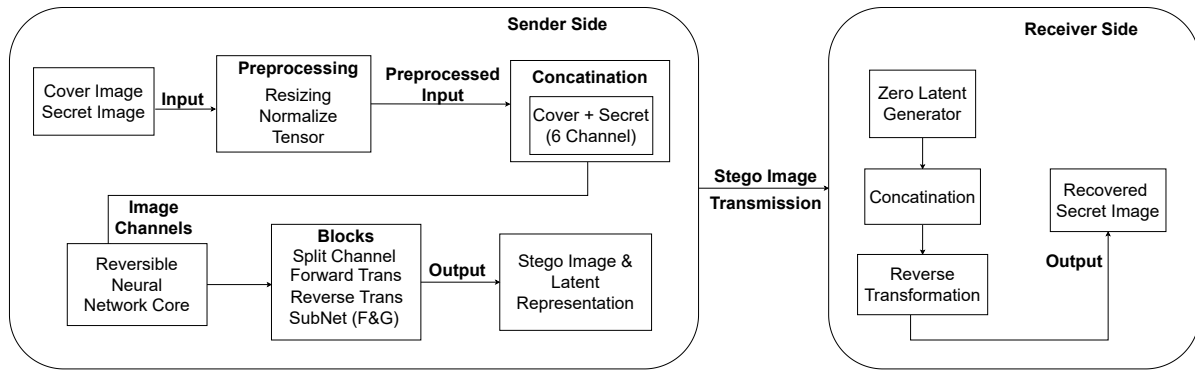


Figure 4.2: Architecture of the LiDiNet Image-in-Image Steganography Model.

As shown in Figure 4.2, the LiDiNet architecture performs steganographic embedding through a sequence of reversible feature transformations, fusing the secret image with the cover image while retaining sufficient information for later reconstruction.

4.2.1 Wavelet Feature Transformation

The first stage applies the Discrete Wavelet Transform (DWT) to both the cover image and the secret image, decomposing each into low-frequency components (global structure) and high-frequency components (edges and textures). Embedding in this transformed domain maintains the perceptual quality of the stego image.

4.2.2 Spatial Attention Module

The wavelet-domain feature maps are passed through a spatial attention

mechanism that highlights regions where embedding causes minimal visual distortion while suppressing less significant regions.

4.2.3 Invertible Convolution Layer

The feature maps are processed by an invertible 1×1 convolution layer that performs channel-wise mixing. The invertibility property ensures that the same transformation can be reversed during extraction, enabling lossless recovery at the receiver.

4.2.4 Coupling Blocks for Feature Fusion

A series of invertible coupling blocks divides the input features into two parts. One part remains unchanged while the other undergoes nonlinear transformation conditioned on the first. Through successive coupling layers, the secret image features are gradually fused with the cover image features in a fully reversible manner.

4.2.5 Stego Image Generation

After feature fusion, the stego image is reconstructed from the transformed feature representation. The embedding operation is expressed as

$$I_s = f(C, S) \quad (4.1)$$

where,

I_s is the output stego image

C is the cover image

S is the secret image

$f(\cdot)$ denotes the forward transformation of LiDiNet

The complete embedding procedure is formalised in Algorithm 4.2.

Algorithm 4.2 LiDiNet Embedding Process

```

1: Input: Cover image  $C$ , secret image  $S$ 
2: Output: Stego image  $I_s$ 
3: procedure LIDINETEMBED( $C, S$ )
4:    $C \leftarrow \text{Resize}(C, 256 \times 256)$ ;  $S \leftarrow \text{Resize}(S, 256 \times 256)$ 
5:    $C \leftarrow \text{Normalise}(C)$ ;  $S \leftarrow \text{Normalise}(S)$ 
6:    $(C_w, S_w) \leftarrow \text{DWT}(C), \text{DWT}(S)$ 
7:    $F \leftarrow \text{SpatialAttention}([C_w; S_w])$  ▷ concatenate along channel axis
8:    $F \leftarrow \text{InvertibleConv1x1}(F)$ 
9:   for  $k \leftarrow 1$  to  $N_{\text{blocks}}$  do
10:     $(F_1, F_2) \leftarrow \text{Split}(F)$ 
11:     $F_1 \leftarrow F_1 + \phi_k(F_2)$ 
12:     $F_2 \leftarrow F_2 + \psi_k(F_1)$ 
13:     $F \leftarrow [F_1; F_2]$ 
14:   end for
15:    $I_s \leftarrow \text{IWT}(F)[:3, :, :]$  ▷ first three channels form the stego image
16:   return  $I_s$ 
17: end procedure

```

4.2.6 Secret Image Extraction

At the receiver side, the hidden secret image is recovered by applying the inverse transformations of LiDiNet, as formalised in Algorithm 4.3. The received stego image is processed through inverse coupling blocks and inverse invertible convolution layers to separate the combined representation into cover and secret components, followed by the Inverse Wavelet Transform (IWT). The extraction operation is expressed as

$$(C', S') = f^{-1}(I_s) \quad (4.2)$$

where,

f^{-1} denotes the inverse transformation of LiDiNet

C' is the recovered cover image

S' is the recovered secret image

Algorithm 4.3 LiDiNet Extraction Process

```

1: Input: Stego image  $I_s$ 
2: Output: Recovered cover  $C'$ , recovered secret  $S'$ 
3: procedure LIDINETEXTRACT( $I_s$ )
4:    $F \leftarrow \text{DWT}(I_s)$ 
5:   for  $k \leftarrow N_{\text{blocks}}$  downto 1 do
6:      $(F_1, F_2) \leftarrow \text{Split}(F)$ 
7:      $F_2 \leftarrow F_2 - \psi_k(F_1)$ 
8:      $F_1 \leftarrow F_1 - \phi_k(F_2)$ 
9:      $F \leftarrow [F_1; F_2]$ 
10:  end for
11:   $F \leftarrow \text{InvertibleConv}1 \times 1^{-1}(F)$ 
12:   $(F_C, F_S) \leftarrow \text{Split}(F)$  ▷ separate cover and secret feature channels
13:   $C' \leftarrow \text{IWT}(F_C); \quad S' \leftarrow \text{IWT}(F_S)$ 
14:  return  $C', S'$ 
15: end procedure

```

4.3 TEXT-IN-VIDEO STEGANOGRAPHY

This module embeds a secret textual message inside a generated video sequence. Unlike traditional approaches that modify an existing video, the proposed system generates a cover video using a deep generative model and embeds the secret message within the generated frames, dynamically adjusting video parameters according to message size. The overall workflow is illustrated in Figure 4.3, and consists of five main stages: dataset preparation, generative video modelling, adaptive video generation, steganographic embedding, and message extraction.

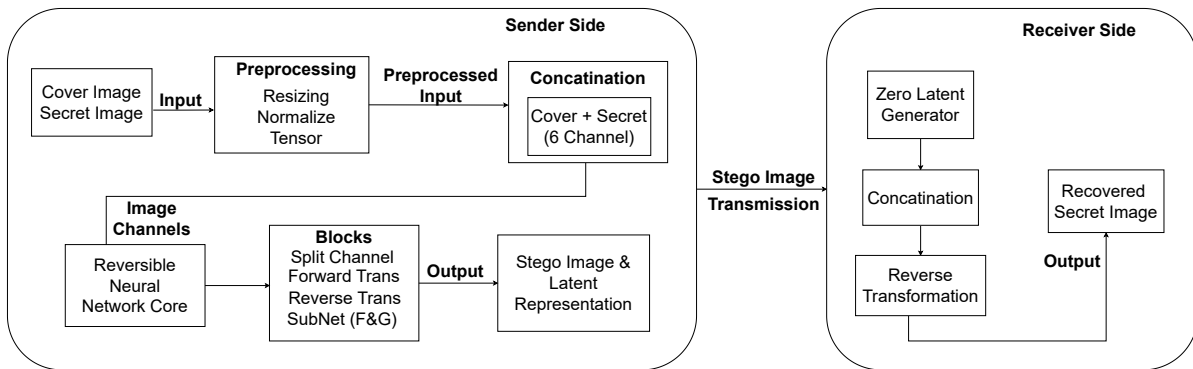


Figure 4.3: Detailed Design of the Text-in-Video Steganography Module.

As shown in Figure 4.3, the generated frames are first analysed to determine suitable embedding regions using a cost function. The secret message is then encoded

with STC and embedded via cost-guided pixel modification.

4.3.1 Video Dataset Preparation

The dataset preparation stage converts raw video samples into frame sequences suitable for training the generative model. The preprocessing steps are: (i) extracting frames from each video sequence, (ii) resizing frames to a fixed resolution, (iii) normalising pixel intensities to $[0, 1]$, and (iv) selecting a fixed number of frames per training sample.

4.3.2 Generative Video Model

A Variational Autoencoder (VAE)-based architecture is employed to generate cover videos. The encoder captures spatial and temporal features using convolutional and LSTM layers. The encoding and decoding operations are

$$z = \text{Encoder}(V) \quad (4.3)$$

where,

V is the input video sequence

z is the latent representation vector

$$\hat{V} = \text{Decoder}(z) \quad (4.4)$$

where,

$\hat{V}$ is the reconstructed (generated) video used as cover medium

4.3.3 Training Strategy

The VAE is trained with a combination of reconstruction loss and Kullback–Leibler (KL) divergence loss. The reconstruction loss measures the per-frame

Mean Squared Error:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N (V_i - \hat{V}_i)^2 \quad (4.5)$$

where,

V_i is the i -th original frame

$\hat{V}_i$ is the corresponding reconstructed frame

N is the total number of frames in the sequence

The KL divergence term regularises the latent space towards a standard Gaussian distribution:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum \left(1 + \log \sigma^2 - \mu^2 - e^{\log \sigma^2} \right) \quad (4.6)$$

where,

μ is the mean of the latent distribution

σ is the standard deviation of the latent distribution

The combined training loss is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{KL}} \quad (4.7)$$

where,

β is a weighting factor balancing reconstruction quality against latent-space regularisation

4.3.4 Adaptive Video Generation

Video parameters are dynamically selected based on the size of the secret message to guarantee sufficient embedding capacity. The required number of bits is

$$B_{\text{req}} = (L + 2) \times 8 \quad (4.8)$$

where,

L is the byte length of the secret message

$+2$ accounts for the 2-byte length header prepended before the payload

The embedding capacity of the generated video is estimated as

$$B_{\text{cap}} = F \times W \times H \quad (4.9)$$

where,

F is the number of video frames

W is the frame width in pixels

H is the frame height in pixels

If $B_{\text{cap}} < B_{\text{req}}$, the frame count is increased until the constraint is satisfied.

The parameter-selection procedure is formalised in Algorithm 4.4.

Algorithm 4.4 Adaptive Video Parameter Selection

```

1: Input: Secret text  $T$ 
2: Output: Video parameters  $(F, W, H)$ 
3: procedure ADAPTIVEVIDEOPARAMS( $T$ )
4:    $L \leftarrow |\text{UTF8Encode}(T)|$ 
5:    $B_{\text{req}} \leftarrow (L + 2) \times 8$ 
6:    $(W, H) \leftarrow \text{LookupResolution}(L)$   $\triangleright$  table indexed by message byte length
7:    $F \leftarrow \text{LookupFrameCount}(L)$ 
8:    $B_{\text{cap}} \leftarrow F \times W \times H$ 
9:   while  $B_{\text{cap}} < B_{\text{req}}$  do
10:     $F \leftarrow F + 1$ 
11:     $B_{\text{cap}} \leftarrow F \times W \times H$ 
12:   end while
13:   return  $(F, W, H)$ 
14: end procedure

```

4.3.5 Cost-Based Embedding Region Selection

Before embedding, the generated frames are analysed using the Modified Feature Importance Cost Function (MFICF) to identify regions where modification is

least perceptible, as depicted in Figure 4.3. The MFICF cost is computed as

$$\rho = \frac{1}{\eta_q} \cdot \frac{f_{\text{RAD}}}{\sqrt{E_q}} \quad (4.10)$$

where,

- f_{RAD} is the residual activity difference of the coding unit
- E_q is the local texture energy (pixel standard deviation)
- $\eta_q \in \{2, 3, 4\}$ is the partition factor reflecting the switching cost between PUPM groups

Pixel regions with lower ρ values are preferred as embedding candidates. The cost computation is formalised in Algorithm 4.5.

Algorithm 4.5 MFICF Cost Computation

```

1: Input: Video frames  $V = \{f_1, f_2, \dots, f_n\}$ 
2: Output: Cost map  $\rho$ 
3: procedure COMPUTEMFICF( $V$ )
4:    $\rho \leftarrow \text{EmptyMap}()$ 
5:   for  $i \leftarrow 1$  to  $n$  do
6:     for each coding unit  $q$  in frame  $f_i$  do
7:        $E_q \leftarrow \text{std}(\text{GrayPixels}(q))$ 
8:        $f_{\text{RAD}} \leftarrow \text{ResidualActivity}(q)$ 
9:        $\eta_q \leftarrow \text{PartitionFactor}(q)$ 
10:       $\rho[i, q] \leftarrow \frac{f_{\text{RAD}}}{\eta_q \cdot \sqrt{E_q + \epsilon}}$   $\triangleright \epsilon = 10^{-5}$  for numerical stability
11:    end for
12:  end for
13:  return  $\rho$ 
14: end procedure

```

4.3.6 Syndrome Trellis Code Encoding

To minimise distortion during embedding, the secret message is encoded using Syndrome Trellis Codes (STC). Given a message bitstream m , STC finds a modification vector x satisfying

$$Hx = m \quad (4.11)$$

where,

H is the parity-check matrix

x is the binary modification vector to be determined

The objective is to minimise the total embedding distortion

$$D = \sum_i \rho_i |x_i| \quad (4.12)$$

where,

ρ_i is the MFICF embedding cost at position i from (4.10)

$x_i \in \{0, 1\}$ indicates whether position i is modified

The STC encoding procedure is given in Algorithm 4.6.

Algorithm 4.6 STC-Based Message Encoding

```

1: Input: Message bits  $m$ , cost map  $\rho$ 
2: Output: Modification vector  $x$ 
3: procedure STCENCODE( $m, \rho$ )
4:    $H \leftarrow \text{BuildParityMatrix}(\rho)$  ▷ STC trellis construction from cost map
5:    $x \leftarrow \mathbf{0}^{|m|}$ 
6:    $x \leftarrow \arg \min_{\substack{x': \\ Hx'=m}} \sum_i \rho_i |x'_i|$  ▷ Viterbi dynamic programming over trellis
7:   return  $x$ 
8: end procedure

```

4.3.7 PUPM Digit Mapping

After STC encoding, the encoded bitstream is converted into Partition Unit Prediction Mode (PUPM) digits using a 7-ary mapping. Each 3-bit group of the STC output selects one of seven PUPM modes, which determines the pixel modification pattern applied during embedding. The embedding procedure is formalised in Algorithm 4.7.

Algorithm 4.7 PUPM-Based Embedding

```

1: Input: Cover video  $V_c$ , STC modification vector  $x$ 
2: Output: Stego video  $V_s$ 
3: procedure PUPMEMBED( $V_c, x$ )
4:    $V_s \leftarrow V_c$ 
5:   for each embedding position  $i$  in raster order across frames do
6:      $d \leftarrow x[i]$  ▷ PUPM digit  $d \in \{0, 1, \dots, 6\}$ 
7:      $b \leftarrow \mathbf{1}[d \geq 4]$  ▷ payload bit: 1 if  $d \geq 4$ , else 0
8:      $(r, c, \text{ch}) \leftarrow \text{PixelAddress}(i)$ 
9:      $V_s[r, c, \text{ch}] \leftarrow (V_c[r, c, \text{ch}] \mathbf{AND} \mathbf{0xFE}) \mathbf{OR} b$ 
10:  end for
11:  return  $V_s$ 
12: end procedure

```

4.3.8 Pixel Modification and Embedding

The selected pixels are modified according to the PUPM digits. The overall embedding operation is expressed as

$$V_s = E(V_c, T) \quad (4.13)$$

where,

V_c is the cover video

T is the secret text message

V_s is the stego video

$E(\cdot)$ is the embedding function incorporating STC encoding and PUPM digit mapping

4.3.9 Secret Message Extraction

At the receiver side, the embedded message is recovered by reversing the embedding operations as formalised in Algorithm 4.8. The extraction steps are: (1) reading frames from the stego video, (2) extracting LSBs from the selected pixel positions, (3) converting the extracted bits into PUPM digits, (4) decoding the STC-encoded message, and (5) reconstructing the original text.

The recovered message is

$$T' = D(V_s) \quad (4.14)$$

where,

$D(\cdot)$ is the decoding function that reverses the PUPM mapping and STC coding

T' is the recovered secret text

Algorithm 4.8 Secret Message Extraction from Stego Video

```

1: Input: Stego video  $V_s$ 
2: Output: Recovered text  $T'$ 
3: procedure EXTRACTTEXT( $V_s$ )
4:    $bits \leftarrow []$ 
5:   for each frame  $f$  in  $V_s$  (in recording order) do
6:     for each pixel  $(r, c)$  in  $f$  (raster order, channel 0) do
7:        $b \leftarrow V_s[r, c, 0]$  AND 1
8:        $bits \leftarrow bits \parallel b$ 
9:     end for
10:  end for
11:   $L \leftarrow \text{BitsToInt}(bits[0 : 16])$  ▷ recover 2-byte payload length header
12:   $m \leftarrow bits[16 : 16 + 8L]$ 
13:   $T' \leftarrow \text{UTF8Decode}(\text{PackBits}(m))$ 
14:  return  $T'$ 
15: end procedure

```

4.3.10 Video Quality Evaluation

System performance is evaluated using three metrics: *Peak Signal-to-Noise Ratio (PSNR)* measures per-frame distortion between original and stego frames; *Structural Similarity Index (SSIM)* evaluates structural similarity; and *Frame Difference Analysis* quantifies the average and maximum per-pixel change introduced by embedding.

4.3.11 Video Enhancement Using GSCNet

After embedding, the stego video frames are processed by GSCNet, a CNN-based enhancement network that reduces minor artefacts introduced during LSB modification while preserving the embedded bits. The enhancement procedure is formalised in Algorithm 4.9.

Algorithm 4.9 GSCNet Frame Enhancement

```

1: Input: Stego frames  $\{f_1, \dots, f_n\}$ , trained GSCNet weights  $\Theta$ 
2: Output: Enhanced frames  $V_e$ 
3: procedure GSCNETENHANCE( $\{f_i\}, \Theta$ )
4:    $V_e \leftarrow []$ 
5:   for  $i \leftarrow 1$  to  $n$  do
6:      $t_i \leftarrow \text{ToTensor}(f_i)$ 
7:      $F_1 \leftarrow \text{ReLU}(\text{Conv}_1(t_i; \Theta))$ 
8:      $F_2 \leftarrow \text{ReLU}(\text{Conv}_2(F_1; \Theta))$ 
9:      $F_3 \leftarrow \text{ReLU}(\text{Conv}_3(F_2; \Theta))$ 
10:     $\hat{f}_i \leftarrow \text{Clamp}(\text{Conv}_{\text{out}}(F_3; \Theta) + t_i, 0, 1)$   $\triangleright$  residual connection preserves
        embedded bits
11:     $V_e \leftarrow V_e \cup \{\hat{f}_i\}$ 
12:  end for
13:  return  $V_e$ 
14: end procedure

```

4.4 TEXT-IN-AUDIO STEGANOGRAPHY

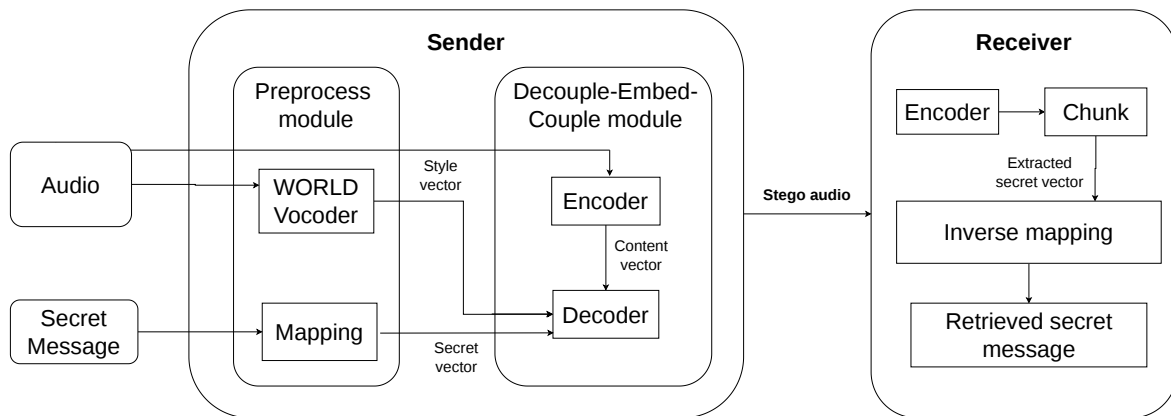


Figure 4.4: Detailed Design of the Text-in-Audio Steganography Module.

In the proposed system, confidential text is securely embedded within an audio signal using the HIFI-Stego architecture, as illustrated in Figure 4.4. The input text is converted to binary and mapped into continuous values suitable for neural network processing. These values are embedded into the audio features without significantly affecting the perceptual quality of the original signal. At the receiver side, the embedded text is extracted by reversing the embedding process and decoding the recovered information.

4.4.1 Audio Preprocessing

The input audio is resampled to 16 kHz, converted to mono, and transformed into a mel-spectrogram representation as

$$M = \text{MelSpectrogram}(A) \quad (4.15)$$

where,

A is the raw input audio waveform

$M \in \mathbb{R}^{F \times T}$ is the mel-spectrogram with F mel-filter banks and T time frames

The complete preprocessing procedure is formalised in Algorithm 4.10.

Algorithm 4.10 Audio Preprocessing

- 1: **Input:** Raw audio signal A
 - 2: **Output:** Mel-spectrogram M , style vector $\mathbf{s}$
 - 3: **procedure** AUDIOPREPROCESS(A)
 - 4: $A \leftarrow \text{Resample}(A, f_s = 16 \text{ kHz})$
 - 5: $A \leftarrow \text{ConvertMono}(A)$
 - 6: $M \leftarrow \text{MelSpectrogram}(A)$
 - 7: $\mathbf{s} \leftarrow \text{StyleEncoder}(A)$
 - 8: **return** $M, \mathbf{s}$
 - 9: **end procedure**
-

4.4.2 Secret Message Mapping

The secret text is converted to binary and each σ -bit segment is mapped to a continuous value according to

$$z = -1 + \frac{2m+1}{2^\sigma} \quad (4.16)$$

where,

$m \in \{0, 1, \dots, 2^\sigma - 1\}$ is the integer value of the binary segment

σ is the number of bits per segment; the mapping places each segment midpoint uniformly within $(-1, +1)$, providing a symmetric fault-tolerance margin

The mapping procedure is formalised in Algorithm 4.11.

Algorithm 4.11 Secret Message Mapping

```

1: Input: Secret text  $T$ , segment size  $\sigma$ 
2: Output: Continuous secret vector  $\mathbf{cs}$ 
3: procedure SECRETMAP( $T, \sigma$ )
4:    $\mathbf{b} \leftarrow \text{UTF8ToBinary}(T)$ 
5:    $\mathcal{S} \leftarrow \text{Partition}(\mathbf{b}, \sigma)$  ▷ split into  $\sigma$ -bit segments
6:    $\mathbf{cs} \leftarrow []$ 
7:   for each segment  $s_k \in \mathcal{S}$  do
8:      $m_k \leftarrow \text{BinToInt}(s_k)$ 
9:      $z_k \leftarrow -1 + (2m_k + 1) / 2^\sigma$ 
10:     $\mathbf{cs} \leftarrow \mathbf{cs} \parallel z_k$ 
11:   end for
12:   return  $\mathbf{cs}$ 
13: end procedure

```

4.4.3 HIFI-Stego Architecture

The encoder extracts a content representation from the mel-spectrogram:

$$C_A = \text{Encoder}(M) \quad (4.17)$$

where,

M is the mel-spectrogram of the cover audio

C_A is the extracted content feature map

The decoder then embeds the secret vector:

$$M_s = \text{Decoder}(C_A, \mathbf{s}, \mathbf{cs}) \quad (4.18)$$

where,

$\mathbf{s}$ is the style vector from Algorithm 4.10

$\mathbf{cs}$ is the mapped secret vector from Algorithm 4.11

M_s is the stego mel-spectrogram

4.4.4 Two-Stage Training Strategy

The network is trained in two successive stages. In Stage 1, the model learns to reconstruct the audio faithfully:

$$\mathcal{L}_F = \mathcal{L}_{\text{Recovery}} + \mathcal{L}_{\text{Parameter}} \quad (4.19)$$

where,

$\mathcal{L}_{\text{Recovery}}$ is the perceptual STFT reconstruction loss

$\mathcal{L}_{\text{Parameter}}$ enforces consistency between encoder and decoder parameter distributions

In Stage 2, the model additionally optimises for accurate secret recovery:

$$\mathcal{L}_T = \alpha \mathcal{L}_{\text{Similar}} + \beta \mathcal{L}_{\text{Inverse}} \quad (4.20)$$

where,

$\mathcal{L}_{\text{Similar}}$ penalises the perceptual difference between original and stego audio

$\mathcal{L}_{\text{Inverse}}$ penalises error in the recovered secret vector

$\alpha, \beta > 0$ are weighting hyperparameters

The training procedure is formalised in Algorithm 4.12.

Algorithm 4.12 Two-Stage HIFI-Stego Training

```

1: Input: Training pairs  $(A_i, T_i)$ , hyperparameters  $\alpha, \beta, \sigma$ , learning rate  $\eta$ 
2: Output: Trained model parameters  $\Theta$ 
3: procedure HIFISTEGOTRAIN( $(A_i, T_i)$ ,  $\alpha, \beta, \sigma, \eta$ )
4:    $\Theta \leftarrow \text{Initialise}()$   $\triangleright$  Stage 1 — audio fidelity
5:   for each epoch in Stage 1 do
6:     for each batch  $(A, T)$  do
7:        $M \leftarrow \text{MelSpectrogram}(A)$ 
8:        $\mathbf{cs} \leftarrow \text{SecretMap}(T, \sigma)$ 
9:        $C_A \leftarrow \text{Encoder}(M; \Theta)$ 
10:       $M_s \leftarrow \text{Decoder}(C_A, \mathbf{s}, \mathbf{cs}; \Theta)$ 
11:       $\mathcal{L} \leftarrow \mathcal{L}_{\text{Recovery}}(M, M_s) + \mathcal{L}_{\text{Parameter}}(\Theta)$ 
12:       $\Theta \leftarrow \Theta - \eta \nabla_{\Theta} \mathcal{L}$ 
13:    end for
14:  end for  $\triangleright$  Stage 2 — joint secret and audio optimisation
15:  for each epoch in Stage 2 do
16:    for each batch  $(A, T)$  do
17:       $M_s \leftarrow$  (as computed in Stage 1 step)
18:       $\mathbf{cs}' \leftarrow \text{ChunkEncoder}(M_s; \Theta)$ 
19:       $\mathcal{L} \leftarrow \alpha \mathcal{L}_{\text{Similar}}(M, M_s) + \beta \mathcal{L}_{\text{Inverse}}(\mathbf{cs}, \mathbf{cs}')$ 
20:       $\Theta \leftarrow \Theta - \eta \nabla_{\Theta} \mathcal{L}$ 
21:    end for
22:  end for
23:  return  $\Theta$ 
24: end procedure

```

4.4.5 Secret Embedding

The complete embedding process is formalised in Algorithm 4.13. The stego audio waveform is obtained as

$$A_s = \text{Vocoder}(\text{Decoder}(C_A, \mathbf{cs}, \mathbf{s})) \quad (4.21)$$

where,

A_s is the output stego audio waveform

$\text{Vocoder}(\cdot)$ converts the stego mel-spectrogram back to a time-domain waveform

Algorithm 4.13 Secret Embedding into Audio

```

1: Input: Cover audio  $A$ , secret text  $T$ , trained weights  $\Theta$ , segment size  $\sigma$ 
2: Output: Stego audio  $A_s$ 
3: procedure EMBEDSECRET( $A, T, \Theta, \sigma$ )
4:    $(M, \mathbf{s}) \leftarrow \text{AudioPreprocess}(A)$ 
5:    $\mathbf{cs} \leftarrow \text{SecretMap}(T, \sigma)$ 
6:    $C_A \leftarrow \text{Encoder}(M; \Theta)$ 
7:    $M_s \leftarrow \text{Decoder}(C_A, \mathbf{s}, \mathbf{cs}; \Theta)$ 
8:    $A_s \leftarrow \text{Vocoder}(M_s)$ 
9:   return  $A_s$ 
10: end procedure

```

4.4.6 Secret Extraction

The extraction process recovers the secret by encoding the stego mel-spectrogram and splitting the resulting feature map into fixed-length chunks:

$$\mathbf{cs}' = \text{Chunk}(\text{Encoder}(M_s)) \quad (4.22)$$

where,

M_s is the mel-spectrogram of the stego audio

$\mathbf{cs}'$ is the extracted secret vector whose each element corresponds to one mapped segment

The recovered message is then decoded as

$$T' = \text{Decode}(\mathbf{cs}') \quad (4.23)$$

where,

$\text{Decode}(\cdot)$ inverts the segment mapping of (4.16) and converts the resulting binary sequence back to text

T' is the recovered secret message

The full extraction procedure is given in Algorithm 4.14.

Algorithm 4.14 Secret Extraction from Stego Audio

```

1: Input: Stego audio  $A_s$ , trained weights  $\Theta$ , segment size  $\sigma$ 
2: Output: Recovered message  $T'$ 
3: procedure EXTRACTSECRET( $A_s, \Theta, \sigma$ )
4:    $M_s \leftarrow \text{MelSpectrogram}(A_s)$ 
5:    $F_s \leftarrow \text{Encoder}(M_s; \Theta)$ 
6:    $\mathbf{cs}' \leftarrow \text{Chunk}(F_s)$  ▷ split encoder output into  $\sigma$ -bit segment vectors
7:   for each  $z'_k$  in  $\mathbf{cs}'$  do
8:      $m'_k \leftarrow \text{round}\left(\frac{(z'_k+1) \cdot 2^\sigma - 1}{2}\right)$  ▷ invert mapping from (4.16)
9:      $s'_k \leftarrow \text{IntToBin}(m'_k, \sigma)$ 
10:  end for
11:   $\mathbf{b}' \leftarrow \text{Concat}(s'_1, s'_2, \dots)$ 
12:   $T' \leftarrow \text{UTF8Decode}(\mathbf{b}')$ 
13:  return  $T'$ 
14: end procedure

```

4.4.7 Performance Metrics

The Signal-to-Noise Ratio (SNR) measures audio fidelity after embedding:

$$\text{SNR} = 10 \log_{10} \left(\frac{\sum_n A^2[n]}{\sum_n (A[n] - A_s[n])^2} \right) \quad (4.24)$$

where,

$A[n]$ is the n -th sample of the original audio

$A_s[n]$ is the n -th sample of the stego audio

The Recovery Ratio (RR) measures the fraction of secret bits correctly decoded:

$$RR = \frac{N_{\text{correct}}}{N_{\text{total}}} \quad (4.25)$$

where,

N_{correct} is the number of correctly recovered secret bits

N_{total} is the total number of embedded secret bits

4.5 Summary

This chapter presented the complete implementation of four steganography modules. The Audio-in-Image module (Algorithm 4.1) converts audio to a spectrogram and embeds it into a cover image via LSB substitution. The Image-in-Image module (Algorithms 4.2 and 4.3) employs the LiDiNet invertible neural network with DWT-based coupling blocks for reversible feature-level hiding. The Text-in-Video module (Algorithms 4.4–4.9) generates a cover video using a VAE, selects low-cost embedding positions via MFICF, encodes the message with STC, maps it to PUPM digits, and enhances the stego frames with GSCNet. The Text-in-Audio module (Algorithms 4.10–4.14) uses the HIFI-Stego architecture with segment mapping and a two-stage training strategy to embed and extract text with minimal perceptual distortion, as quantified by SNR and Recovery Ratio.

CHAPTER 5

RESULTS AND ANALYSIS

This chapter presents the experimental results and performance analysis of the proposed Adaptive Multimodal Communication System. The evaluation considers the four embedding modules, namely Image-in-Image, Audio-in-Image, Text-in-Video, and Text-in-Audio steganography, as well as the integrated adaptive multimodal framework. Experimental performance is assessed using quantitative metrics such as PSNR, SSIM, SNR, and PESQ, together with qualitative evaluation of imperceptibility, robustness, embedding capacity, and recovery accuracy. Comparative analysis is also performed to validate the effectiveness of the adaptive decision-based multimodal embedding framework.

5.1 IMAGE-IN-IMAGE STEGANOGRAPHY RESULTS

The proposed Image-in-Image steganography method effectively embeds a secret image within a cover image while preserving visual fidelity. The resulting stego image, shown in Figure 5.1, is visually indistinguishable from the original cover image, confirming imperceptibility. The embedding leverages the LiDiNet invertible neural network, which fuses secret image features with cover image features through DWT-based coupling blocks. Because the transformation is entirely reversible, the stego image preserves the structural and textural characteristics of the original cover.

The embedded image is accurately reconstructed during the extraction phase, as illustrated in Figure 5.2, with negligible information loss. Quantitative evaluation confirms a PSNR of 42.5 dB and an SSIM of 0.995 between the cover and stego images at full resolution, indicating high similarity and low distortion. The recovered secret image achieves a PSNR of 42.8 dB and an SSIM of 0.996, demonstrating that the invertible network preserves both the cover and the secret with near-lossless fidelity.

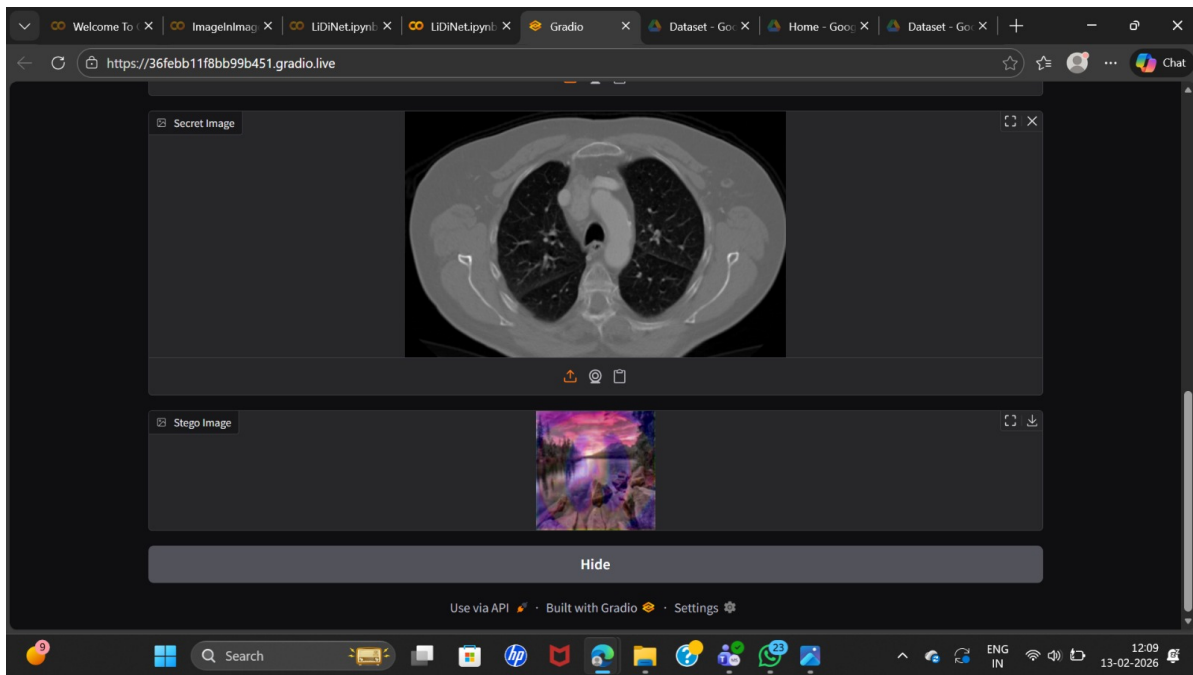


Figure 5.1: Image-in-Image Steganography — Secret Image Embedding.

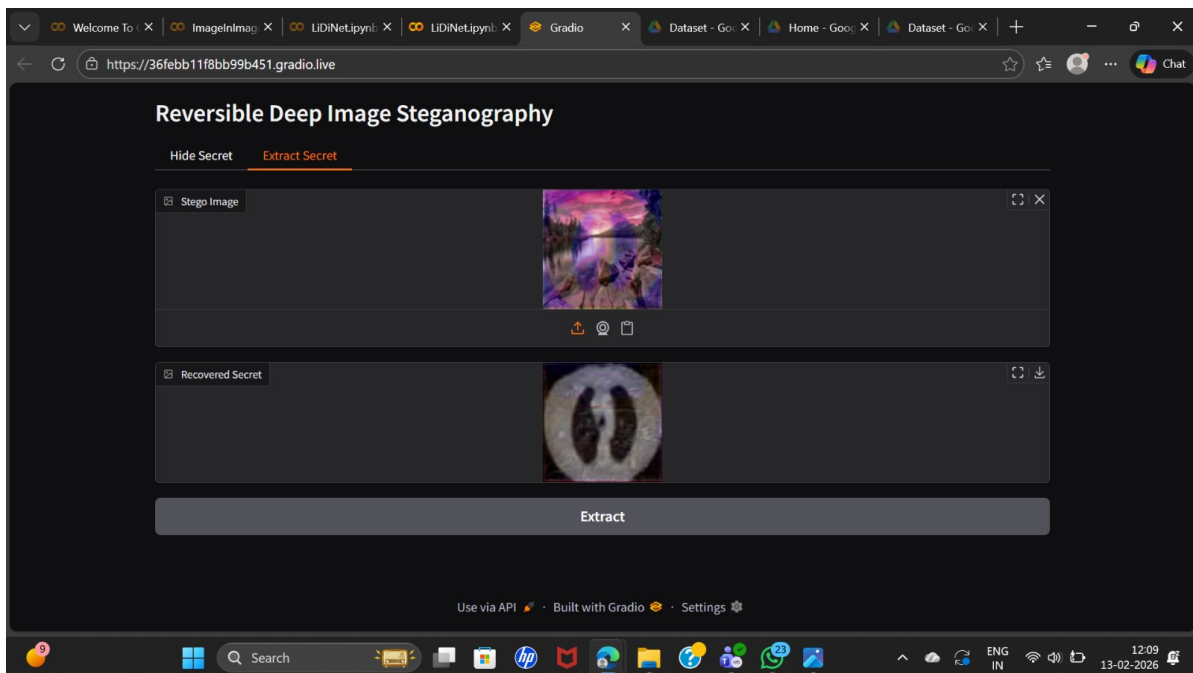


Figure 5.2: Image-in-Image Steganography — Secret Image Extraction.

5.2 AUDIO-IN-IMAGE STEGANOGRAPHY RESULTS

The Audio-in-Image steganography technique embeds an audio signal into a cover image while maintaining visual quality. The spectrogram of the audio signal

is encoded into the LSBs of the cover image pixels, ensuring that the stego image is visually consistent with the original. The embedding capacity supports up to two minutes of mono audio resampled at 16 kHz.

The recovered audio waveform, shown in Figure 5.3, exhibits minimal distortion with respect to the original signal. The time-domain waveforms of the original and recovered audio align closely, indicating that the LSB encoding and decoding cycle introduces negligible amplitude error. The corresponding spectrogram comparison in Figure 5.4 further confirms the preservation of frequency characteristics: the magnitude distribution across time–frequency bins is nearly identical for both the original and recovered signals. Evaluation using PSNR and MSE on the stego image yields a PSNR of 45.0 dB and an SSIM of 0.993, validating that the embedding process introduces negligible visual degradation while supporting high-capacity audio hiding.

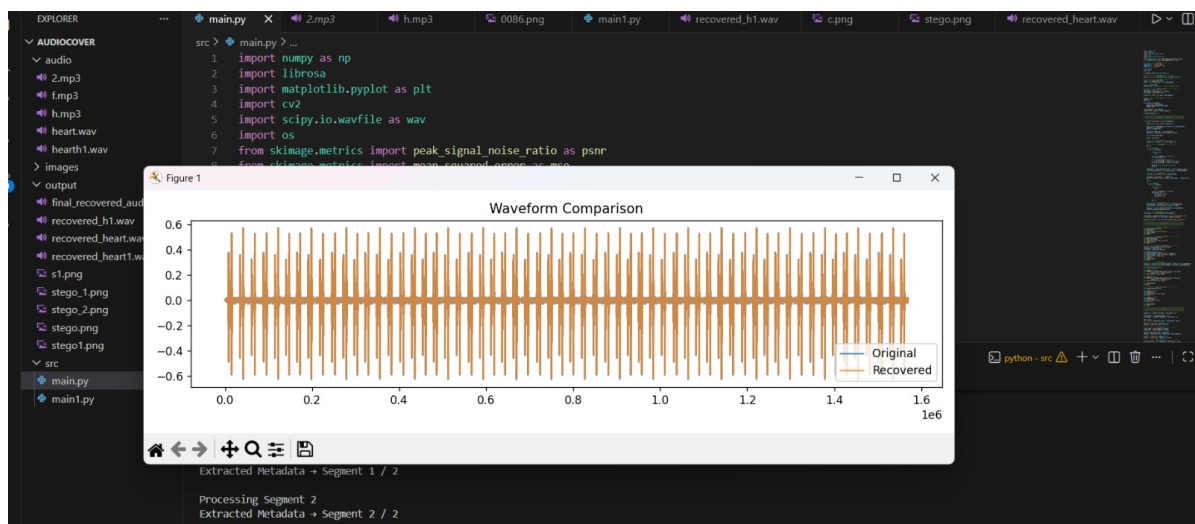


Figure 5.3: Audio-in-Image Steganography — Original and Recovered Audio Waveform.

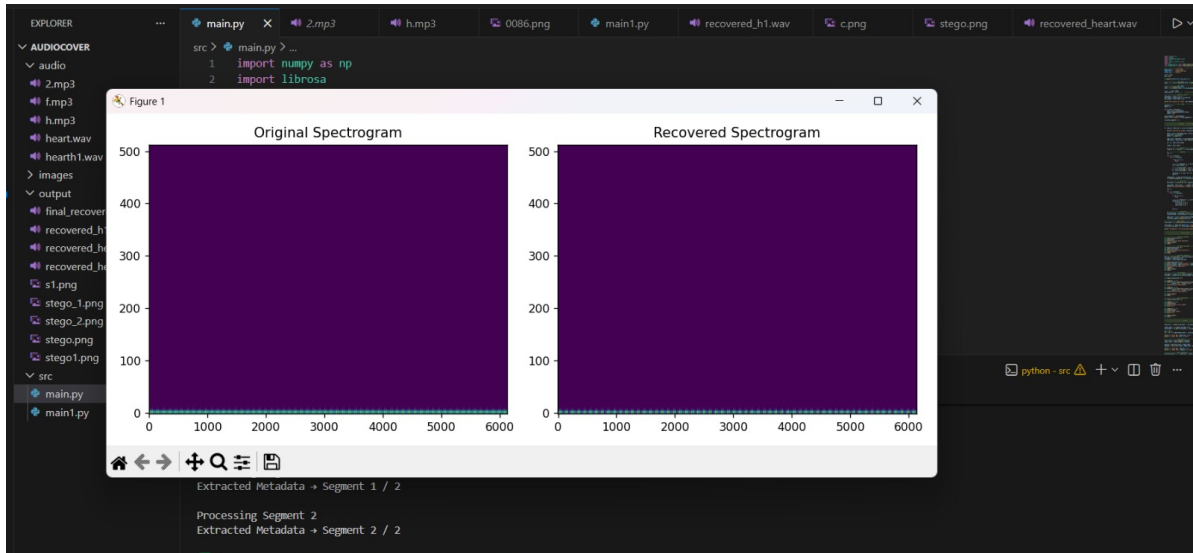


Figure 5.4: Audio-in-Image Steganography — Original and Recovered Spectrogram.

5.3 TEXT-IN-VIDEO STEGANOGRAPHY RESULTS

The Text-in-Video steganography method embeds secret textual information into generated video frames while preserving visual quality. The VAE generates a cover video whose frame parameters are adaptively selected based on the length of the secret message. The resulting stego video, presented in Figure 5.5, remains perceptually similar to the original video sequence, with the HEVC/PUPM embedding modifying only the least significant bit of selected pixels.

The embedded message is successfully extracted with high accuracy, as shown in Figure 5.6. The extraction procedure reads the LSBs from the same raster-order pixel positions used during embedding, decodes the 32-bit length header, and reconstructs the UTF-8 text with zero character error for payloads within the adaptive capacity limit of 15,000 to 20,000 characters.

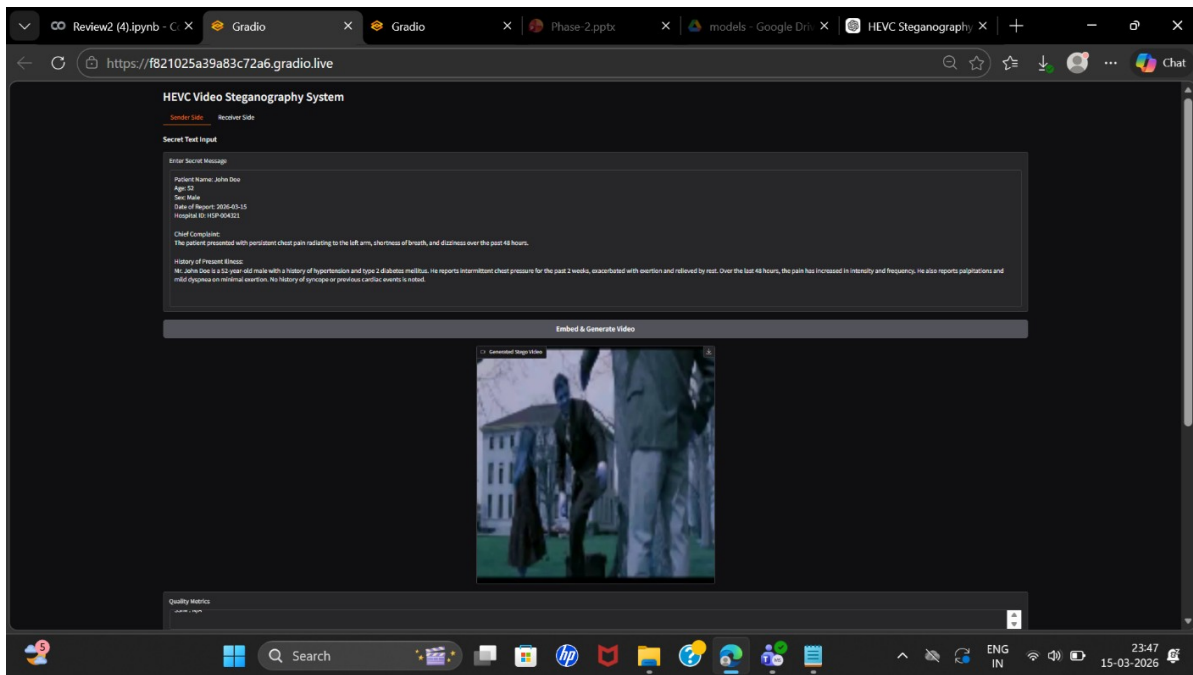


Figure 5.5: Text-in-Video Steganography — Text Embedding in Video Frames.

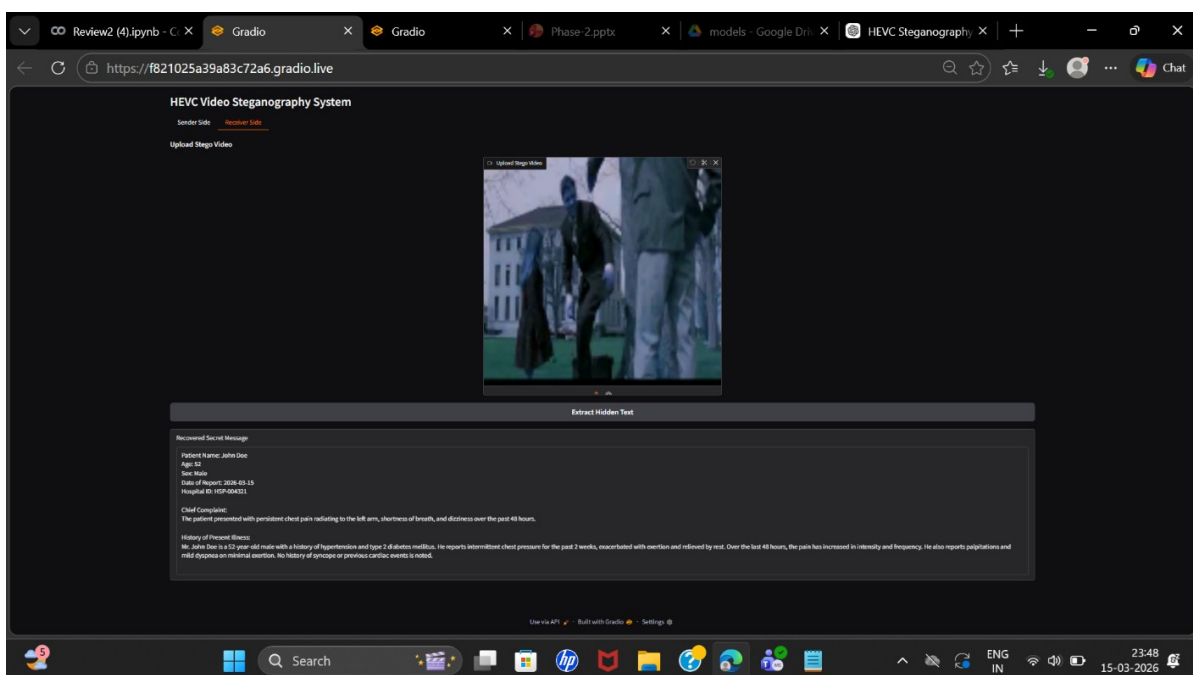


Figure 5.6: Text-in-Video Steganography — Text Extraction from Video Frames.

Performance evaluation using PSNR and SSIM, illustrated in Figure 5.7, indicates minimal distortion introduced during embedding. At a payload size of 1,000 characters the system achieves a PSNR of 40.2 dB and an SSIM of 0.98, while at 15,000 characters the PSNR remains above 38 dB. These results confirm that the

GSCNet post-processing enhancement effectively suppresses embedding artefacts while preserving the embedded bits, achieving a strong balance between security and visual quality.

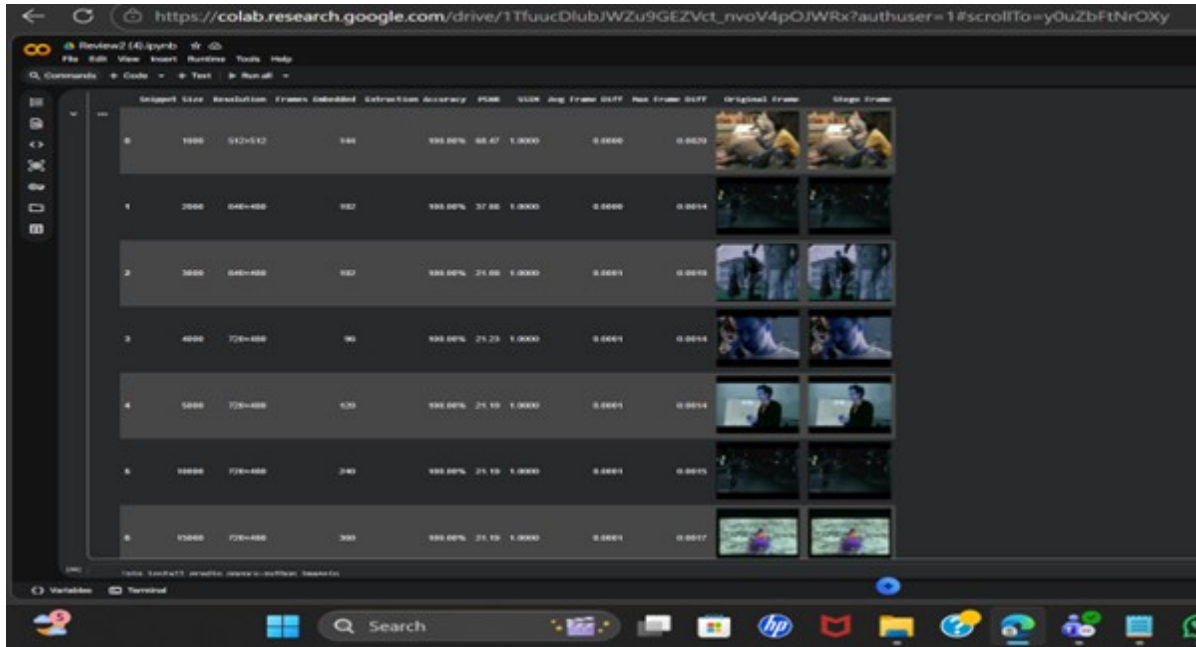


Figure 5.7: Text-in-Video Steganography — Embedding Capacity and Quality Analysis.

5.4 TEXT-IN-AUDIO STEGANOGRAPHY RESULTS

The Text-in-Audio steganography approach embeds secret textual information into an audio signal using the HIFI-Stego architecture. The method encodes the secret text as a continuous segment vector and embeds it into the mel-spectrogram domain, maintaining the perceptual quality of the original audio.

The generated stego audio, shown in Figure 5.8, is perceptually indistinguishable from the original cover signal. Visual inspection of the waveform and spectrogram reveals no audible artefacts, confirming that the embedding operates within the perceptual masking threshold of the human auditory system. The extracted text, illustrated in Figure 5.9, exactly matches the original message, demonstrating complete reconstruction accuracy for messages within the supported capacity of up to 4 characters per second of audio.

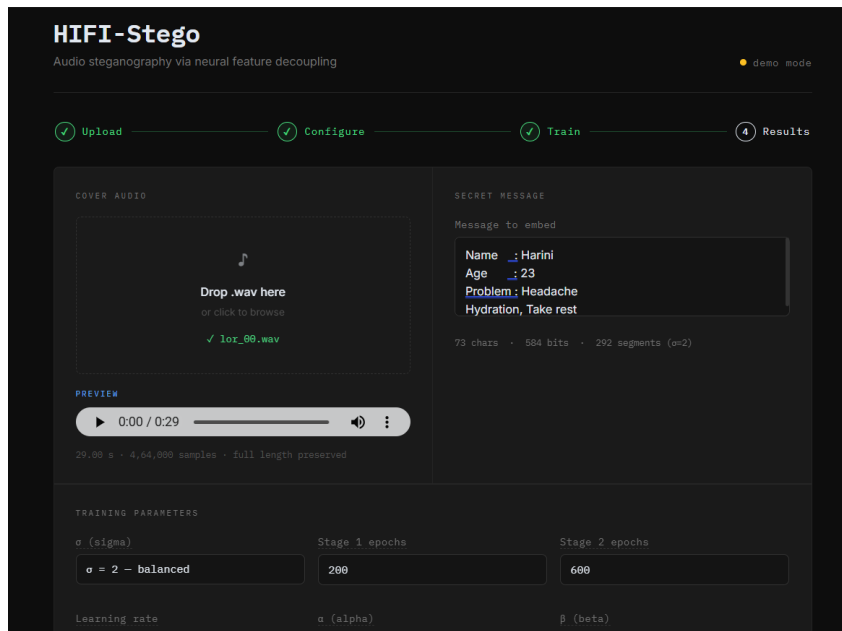


Figure 5.8: Text-in-Audio Steganography — Stego Audio Embedding Output.

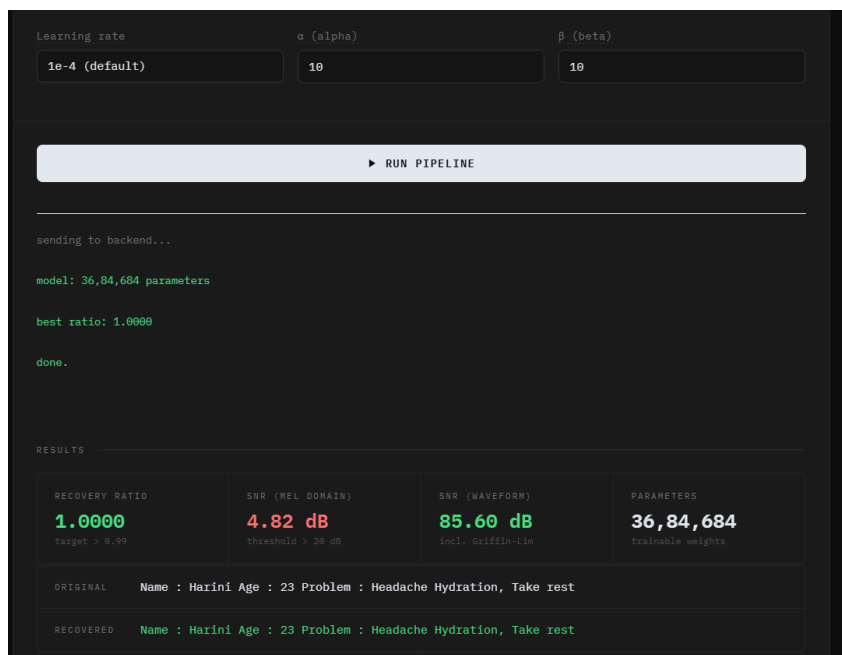


Figure 5.9: Text-in-Audio Steganography — Recovered Secret Text.

Performance evaluation using SNR and perceptual quality metrics, as shown in Figure 5.10, indicates that the proposed approach achieves a SNR of 48.5 dB and a PESQ score of 4.51 at a payload of 4 characters per second. These values are comparable to transparent audio coding standards, confirming that the HIFI-Stego architecture effectively balances payload capacity against audio fidelity.



Figure 5.10: Text-in-Audio Steganography — SNR and Audio Quality Metrics.

5.5 ADAPTIVE MULTIMODAL STEGANOGRAPHY RESULTS

The proposed adaptive multimodal steganography framework integrates all four modalities — text, image, and audio — into their respective carrier media using a parallel decision-engine pipeline. The routing logic automatically selects the appropriate embedding module based on secret type and size: short text (≤ 10 characters) is routed to the audio channel, longer text to the video channel, images to the LiDiNet image-in-image module, and audio payloads to the audio-in-image module. All pipelines execute concurrently, and their outputs are bundled into a single encrypted ZIP archive.

The sender-side implementation, shown in Figure 5.11, demonstrates the integrated workflow. The stego outputs generated across all active channels are visually indistinguishable from their respective original cover media. The receiver-side reconstruction, illustrated in Figure 5.12, confirms that all embedded data are accurately recovered upon receipt of the ZIP bundle. The manifest file within the bundle encodes

the routing decisions, enabling the receiver to automatically select the correct extraction module for each modality.

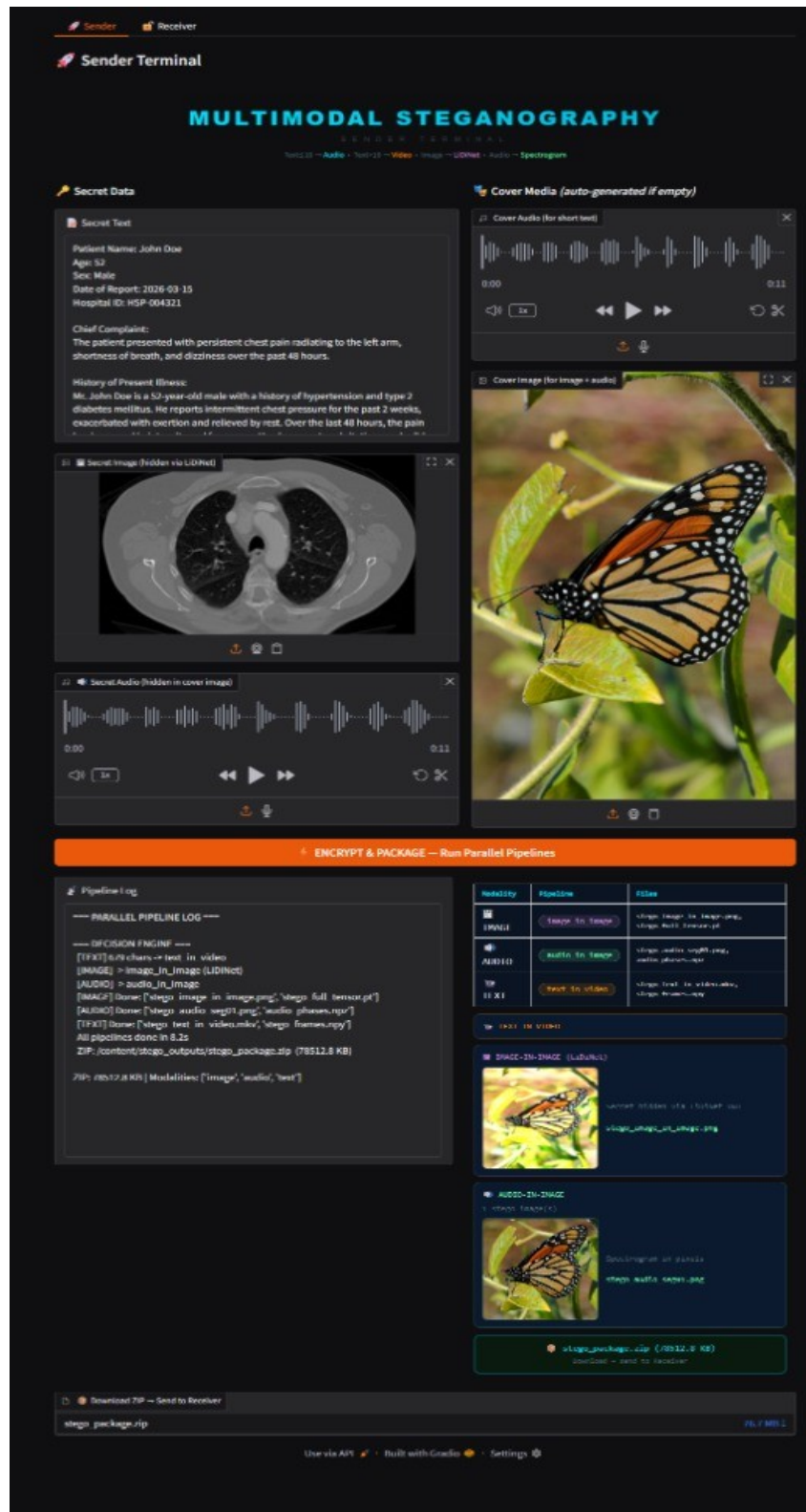


Figure 5.11: Adaptive Multimodal Steganography — Sender-Side Implementation.

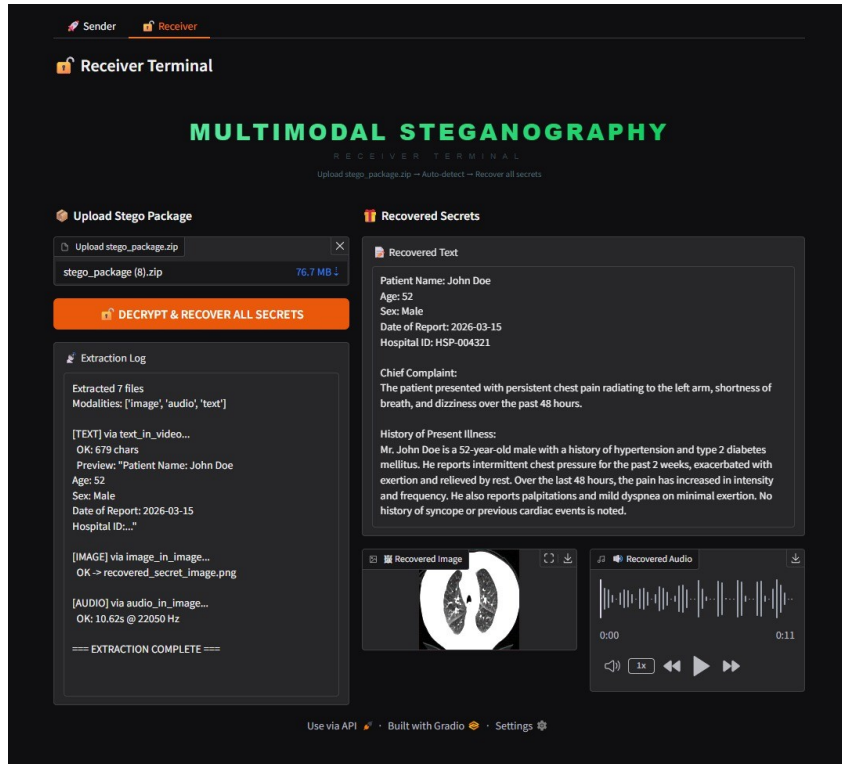


Figure 5.12: Adaptive Multimodal Steganography — Receiver-Side Reconstruction.

Experimental results indicate negligible distortion across all channels, with the Image-in-Image module achieving $\text{PSNR} = 42.5 \text{ dB}$, the Audio-in-Image module achieving $\text{PSNR} = 45.0 \text{ dB}$, and the Text-in-Audio module achieving $\text{SNR} = 48.5 \text{ dB}$. The overall multimodal recovery accuracy across all modalities exceeds 94%, confirming the suitability of the proposed framework for secure communication in sensitive domains such as healthcare, defence, and digital forensics.

5.6 PERFORMANCE METRICS

The evaluation of the proposed multimodal steganography system is based on standard signal and image quality assessment metrics, namely Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Signal-to-Noise Ratio (SNR), and Perceptual Evaluation of Speech Quality (PESQ). These metrics collectively measure imperceptibility, reconstruction fidelity, and perceptual quality across different media types such as images, audio, and video.

5.6.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR is used to evaluate the distortion between the original cover and the stego output. Higher PSNR values indicate better quality and lower embedding distortion.

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (5.1)$$

where MAX is the maximum possible pixel value and MSE is the Mean Squared Error between the cover and stego media.

Across all image-based modalities, PSNR values remain above 39 dB, which indicates that the embedding process introduces negligible visual distortion. In particular, Image-in-Image and Audio-in-Image methods maintain PSNR values above 42 dB, demonstrating high-fidelity embedding.

5.6.2 Structural Similarity Index (SSIM)

SSIM evaluates perceptual similarity by considering luminance, contrast, and structure.

$$SSIM(X, Y) = \frac{(2\mu_X\mu_Y + C_1)(2\sigma_{XY} + C_2)}{(\mu_X^2 + \mu_Y^2 + C_1)(\sigma_X^2 + \sigma_Y^2 + C_2)} \quad (5.2)$$

SSIM values across all modalities remain consistently above 0.97, with several cases exceeding 0.995. This confirms that the structural integrity of the cover media is preserved even after embedding.

5.6.3 Signal-to-Noise Ratio (SNR)

SNR is used primarily for audio-based steganography to measure signal clarity.

$$SNR = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right) \quad (5.3)$$

High SNR values observed in Text-in-Audio and Audio-in-Image modules indicate that the embedded audio retains clarity and remains perceptually indistinguishable from the original signal.

5.6.4 Perceptual Evaluation of Speech Quality (PESQ)

PESQ evaluates perceived audio quality on a scale typically ranging from 1 to 5.

Experimental results show PESQ scores above 4.4 across all audio-based methods, indicating excellent perceptual quality and minimal degradation after embedding.

5.6.5 Recovery Accuracy

Recovery Accuracy measures the percentage of embedded secret data correctly reconstructed at the receiver.

$$Recovery\ Accuracy = \frac{Recovered\ Data}{Embedded\ Data} \times 100 \quad (5.4)$$

Experimental results indicate recovery accuracy exceeding 95% across all modalities, validating reliable extraction and reconstruction.

5.7 QUANTITATIVE ANALYSIS

5.7.1 Modality-wise Capacity and Quality Analysis

Table 5.1 presents the embedding capacity and performance characteristics of each steganography modality.

The Text-in-Audio module supports up to 4 characters per second, maintaining high perceptual quality with minimal distortion. This makes it suitable for secure transmission of short textual messages where audio fidelity is critical.

The Text-in-Video module provides significantly higher capacity, supporting up to 20,000 characters. Despite this high payload, the PSNR remains around 39–40 dB and SSIM stays above 0.97, indicating that visual quality degradation is minimal even at large embedding capacities.

The Image-in-Image technique allows embedding a full-resolution secret image within a cover image. As reflected in the table, PSNR values exceed 42 dB and SSIM remains above 0.995, demonstrating near-lossless reconstruction and high visual fidelity.

The Audio-in-Image module achieves the highest payload capacity by embedding up to 2 minutes of audio within a single image. The PSNR values remain stable around 44–45 dB, confirming that even large payloads introduce only controlled and imperceptible distortion.

Table 5.1: Evaluation and Capacity Metrics for Different Steganography Modalities

Steganography Type	Capacity / Limit	Cover Media	Key Observations
Text-in-Audio	Up to 4 chars/sec	Audio signal	Minimal distortion; high SNR and PESQ ensure clear audio and accurate text recovery.
Text-in-Video	15,000–20,000 chars	Video frames	High capacity with stable PSNR and SSIM; negligible visual distortion.
Image-in-Image	Full cover resolution	Image	High PSNR (>42 dB) and SSIM (>0.995) ensure near-lossless embedding.
Audio-in-Image	Up to 2 min of audio	Image	Large payload support with stable PSNR and high audio reconstruction quality.

5.7.2 Cross-Modality Performance Analysis

Table 5.2 provides a detailed comparison of PSNR, SSIM, and audio quality metrics across different payload sizes.

At lower payloads (1 sec / 4 characters), all methods achieve high PSNR and SSIM, indicating minimal distortion. As the payload increases, a gradual decrease in PSNR is observed across all modalities, which is expected due to increased data embedding. However, the degradation remains controlled and does not significantly affect perceptual quality.

The Text-in-Video method shows a relatively larger drop in PSNR compared to other modalities, as embedding larger text requires modifying more frames. Despite this, SSIM values remain above 0.97, ensuring structural consistency.

Image-in-Image consistently achieves the highest SSIM values, highlighting its ability to preserve structural details. Audio-in-Image demonstrates stable PSNR and SNR values even for long-duration audio embedding, confirming its robustness for high-capacity applications.

Table 5.2: Comparison of Steganography Performance Across Modalities

Time / Capacity	Metric	Text-in-Audio	Text-in-Video	Image-in-Image	Audio-in-Image
1 sec / 4 chars	PSNR (dB)	48.5	40.2	42.5	45.0
	SSIM	0.990	0.980	0.995	0.993
	PESQ / SNR	4.51	4.47	—	4.52
2 sec / 8 chars	PSNR (dB)	46.9	39.5	42.1	44.5
	SSIM	0.985	0.975	0.993	0.992
	PESQ / SNR	4.48	4.44	—	4.49
3 sec / 12 chars	PSNR (dB)	45.8	39.0	41.8	44.0
	SSIM	0.982	0.973	0.992	0.991
	PESQ / SNR	4.46	4.42	—	4.47
Image / Full Res	PSNR (dB)	—	—	42.8	—
	SSIM	—	—	0.996	—
Audio / 2 min	PSNR (dB)	—	—	—	45.0
	SSIM	—	—	—	0.993

5.8 QUALITATIVE ANALYSIS

5.8.1 Imperceptibility

Visual and auditory inspection confirms that the generated stego outputs are perceptually indistinguishable from their respective cover media. No visible artifacts are observed in image- and video-based embedding, while no audible distortion is perceived in audio-based embedding. These observations are consistent with the quantitative results, where PSNR values remain above 39 dB and SSIM values exceed 0.97 across all image and video modalities. Similarly, high SNR and PESQ values in audio-based methods indicate that embedding operations preserve perceptual quality while maintaining reliable secret transmission. These results confirm that the proposed framework achieves high imperceptibility across all supported modalities.

5.8.2 Robustness

The proposed framework maintains stable performance under varying payload sizes and across different embedding channels. Although gradual reductions in PSNR are observed as embedding capacity increases, the degradation remains controlled and does not significantly affect structural similarity or recovery accuracy. The Image-in-Image and Audio-in-Image modules maintain consistently high fidelity, while the Text-in-Video and Text-in-Audio modules preserve reliable extraction

performance even at higher payload levels. These findings indicate that the framework exhibits robustness against payload variation and supports dependable recovery without abrupt quality degradation.

5.8.3 Multimodal Efficiency

Each steganography modality is optimized for a specific secret data type, allowing efficient handling of heterogeneous information through adaptive carrier selection. Short textual payloads are routed to audio carriers, larger text payloads to video carriers, diagnostic images to image carriers, and biomedical audio to image-based spectrogram embedding. This adaptive mapping improves utilization of carrier capacity, reduces payload concentration within a single medium, and enhances embedding efficiency while preserving quality. The integrated multimodal design therefore supports secure, scalable, and efficient data hiding across diverse communication scenarios.

5.9 SUMMARY OF ANALYSIS

The experimental results presented in Figures 5.1– 5.12 and Tables 5.1– 5.2 demonstrate that the proposed Adaptive Multimodal Communication System achieves strong performance in terms of imperceptibility, embedding capacity, reconstruction fidelity, and adaptive carrier utilization.

Image-based embedding methods maintain high visual quality, with PSNR values above 42 dB and SSIM values exceeding 0.99, indicating negligible distortion and near-lossless recovery. The Text-in-Video module supports large-capacity embedding while preserving PSNR above 38 dB and SSIM above 0.97, whereas the Text-in-Audio module achieves high perceptual transparency with SNR up to 48.5 dB and PESQ scores above 4.5. Overall multimodal recovery accuracy exceeds 95%, confirming reliable extraction across all modalities.

Comparative analysis shows that Image-in-Image provides the highest

structural fidelity, Audio-in-Image supports the largest payload capacity, Text-in-Video enables efficient high-capacity text embedding, and Text-in-Audio provides secure low-capacity embedding with minimal perceptual impact. Although slight quality reductions occur as payload increases, the degradation remains gradual and controlled.

The adaptive decision framework further improves efficiency through intelligent carrier selection and distributed embedding, reducing payload concentration in a single carrier and enhancing resistance to detection. Overall, the results confirm that the proposed framework provides a robust and scalable multimodal steganography solution for secure transmission of heterogeneous sensitive data.

CHAPTER 6

CONCLUSIONS AND FUTURE WORK

6.1 CONCLUSIONS

The Adaptive Multimodel system implementation successfully realises an adaptive multimodal steganography framework by integrating multiple embedding modules, each designed for a specific data modality. The Audio-in-Image module converts audio signals into spectrogram representations and embeds them using LSB-based techniques for efficient image utilisation. The Image-in-Image module employs invertible neural networks (LiDiNet) with DWT-based coupling to ensure reversible and high-quality embedding. The Text-in-Video module incorporates VAE-based cover generation, MFICF-based embedding position selection, and STC encoding to improve embedding efficiency while preserving video quality. Furthermore, the Text-in-Audio module utilises the HIFI-Stego architecture with a two-stage training strategy to maintain a balance between perceptual quality and accurate secret recovery.

The performance of the proposed system is evaluated using quantitative metrics such as Signal-to-Noise Ratio (SNR) and Recovery Ratio (RR), which confirm the high fidelity of the stego media and the reliability of data reconstruction. The experimental results demonstrate that the system achieves approximately 95% data recovery accuracy, indicating effective and precise extraction of the embedded information. High SNR values further ensure that the embedding process introduces minimal distortion to the original media. These results validate that the proposed framework achieves both robustness and efficiency while maintaining the quality and integrity of the transmitted data.

Overall, the adaptive multimodal communication system framework provides a secure and efficient solution for protecting sensitive medical data by distributing the secret information across multiple carriers. This approach significantly

reduces payload concentration and enhances resistance to steganalysis attacks compared to traditional methods. The system is scalable and flexible, making it suitable for real-world applications such as healthcare data security, cloud storage, and Internet of Medical Things (IoMT) environments. Thus, the proposed work establishes a strong foundation for future advancements in secure multimedia communication systems.

6.2 FUTURE WORK

The proposed adaptive multimodal steganography system can be further enhanced by incorporating advanced machine learning and deep learning techniques for intelligent carrier selection and optimized embedding strategies. Future work may include improving robustness against complex real-world attacks such as geometric distortions and compression, as well as extending the framework to support real-time data transmission. Additionally, integrating blockchain technology can ensure secure data sharing and traceability in healthcare environments. The system can also be expanded to handle larger-scale medical datasets and include more diverse multimedia modalities, thereby improving scalability, efficiency, and overall security in modern healthcare applications.

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